

MULTIPLE CHOICE QUESTIONS

1. The study of physics deals with:

A The Laws of motion	C The forces present in the universe
B The structure of space and time	D All of the above
2. The branch of science which deals with the study of matter and energy and the relationship between them is called:

A Astronomy	B Geology	C Physics	D Biology
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3. The system international (SI) is built up from:

A Base units	C Derived units
B Supplementary units	D All of the above
4. Which of the following Muslim scientist is both poet and mathematician?

A Omar- Khayyam	B Ibn -e- Sina	C Al – Khwarizmi	D Al-Kindi
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5. Which Muslim scientist first invented logarithm?

A Omar – Khayyam	C Ibn – e –Sina
B Al – Khwarizmi	D Al – Beruni
6. Al – Khwarizmi's Hisabul – Jubr – WAP – Muqabla was based on which subject:

A Analytical Algebra	C Trigonometry
B Geometry	D Philosophy
7. "Kitab-ul- Qanoon- ul- Masoodi" is the famous book of:

A Ibn-ul- Haithem	C Al- Beruni
B Al- Kindi	D Al- Khwarizmi
8. Al- Shifa was written by:

A Al- Beruni	C Al-Khwarizmi
B Jabir-Bin-Hayyan	D Ibn-e-Sina
9. Pin-hole camera was invented by:

A Al-Beruni	C Al-Khwarizmi
B Ibn-ul-Haithem	D Al-Kindi
10. Al-Manazir is the famous book of:

A Al-Beruni	B Al-Kindi	C Ibn-ul-Haithem	D Al-Khwarizmi
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11. The quantities like length, time, mass, force, electric charge and many more are called:

A Basic quantities	C Derived quantities
B Physical quantities	D Specified quantities
12. Quantities like length, mass, time, temperature, electric current and intensity of light are:

A Basic quantities	C Derived quantities
B Physical quantities	D Specified quantities
13. All quantities which can be described in terms of basic quantities are:

A Basic quantities	C Derived quantities
B Physical quantities	D Specified quantities
14. The S.I units consists of:

A Five basic units	C Seven Derived
B Five derive units	D Seven basic units

15. The S.I unit of electric current is:
 A Mole B Ampere C Candela D Kelvin
16. The S.I unit of mass is:
 A Kg B gm C IBS D Slug
17. The S.I unit of temperature is:
 A Mole B Ampere C Candela D Kelvin
18. The S.I unit of amount of substance is:
 A Meter B Kilogram C Candela D Mole
19. The S.I unit of intensity of light is:
 A Meter B Kilogram C Candela D Mole
20. Which of the following is not a unit of energy?
 A Kilowatt B Erg C Joule D Kilowatt-hour
21. Erg x sec is the unit of:
 A Angular momentum C Planck's constant
 B Linear momentum D Energy
22. Newton second is a unit of:
 A Energy B Momentum C Force D Work
23. The S.I unit of pressure is:
 A Atmosphere B Dyne/cm² C co of Hg D Pascal
24. Joule-second is a unit of:
 A Energy C Power
 B Angular momentum D Momentum
25. Which of the following is (are) dimensionless?
 A Refractive index C Universal gravitation constant
 B Specific heat D Momentum
26. The dimensional formula for angular velocity is:
 A $M L^0 T^2$ B $M L T^1$ C $M^0 L^0 T^1$ D $M^0 L^0 T^0$
27. The dimensional formula for μ_0 is:
 A $M L^3 T^2$ B $M^{-1} L^3 T^2$ C $M^{-1} L^3 T^2$ D $M L^2 T^3$
28. The dimensional formula for potential difference is:
 A $M L^2 T^3 I^{-1}$ B $M^2 L T^3 I$ C $M L^2 T^2 I^{-1}$ D $M L T^3 I^{-1}$
29. The dimensional formula for resistance is:
 A $M L^2 T^3 I^{-1}$ B $M L^2 T^2 I^{-2}$ C $M L^2 T^3 I^2$ D $M L T^3 I^2$
30. The dimensional formula for inductance is:
 A $M L^2 T^2 I^{-1}$ B $M L^2 T^2 I^{-2}$ C $M L^{-2} T^2 I^2$ D $M^2 L^2 T^3 I^{-2}$

CHAPTER 02 SCALAR & VECTORS

MULTIPLE CHOICE QUESTIONS

1. If the resultant of two vectors each of magnitudes F is also of magnitude F , the angle b/w them will be:
 A 90 B 60 C 30 D 120
2. If $|A + B| = |A - B|$, the angle b/w the vectors is:
 A 0° B 90° C 45° D 120°
3. The resultant of two forces of 3N and 4N making an angle 90° with each other is:
 A 1.0 N B 7 N C 5 N D 3.5 N
4. Which pair of the following forces will never give resultant force of 2N?
 A 2N, 2N B 1N, 1N C 1N, 3N D 1N, 4N
5. If the resultant of two vectors each of magnitudes F , is twice of magnitude F , the angle b/w them will be:
 A 90° B 60° C 0° D 120°
6. Two forces each of 10N magnitude acting on a body, if the forces are inclined at 30° and 60° with x-axis, the x- component of their resultant is:
 A 10 N B 136.6 N C 1.366 N D 13.66 N
7. The position vector of a particle is given by $\vec{r} = 3t^2\hat{i} + 4t^2\hat{j} + 7\hat{k}$. Find the displacement after 10 seconds. (LUMS 2007)
 A 500 m B 650 m C 350 m D 400 m
8. A force $F = 2\hat{i} + 3\hat{j}$ units, has its point of application moved from the point (1,3) to the point (5,7), work done is:
 A 20 B 30 C 40 D 50
9. The angle b/w the vector $A = 2\hat{i} + 3\hat{j} - \hat{k}$ and $B = 4\hat{i} + 6\hat{j} - 2\hat{k}$, is:
 A 0° B 60° C 90° D 180°
10. The angle b/w the vector $A = 2\hat{i} + 3\hat{j} + \hat{k}$ and $B = -6\hat{i} - 9\hat{j} + 3\hat{k}$, is:
 A 0° B 60° C 90° D 180°
11. The angle b/w the vector $A = 2\hat{i} + 3\hat{j} + 10\hat{k}$ and $B = -6\hat{i} - 9\hat{j} + 3\hat{k}$, is:
 A 0° B 60° C 90° D 180°
12. The angle b/w the vector $A = \hat{i} + \hat{j}$ and $B = \hat{j} + \hat{k}$, is:
 A 0° B 60° C 90° D 180°
13. If the vector A and B are of magnitude 4 and 3 cm making of 30° and 90° respectively with x-axis, their scalar product will be:
 A 0 cm^2 B 18 cm^2 C 6.0 cm^2 D 21 cm^2
14. A particle moves from position $r_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $r_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of a force $F = 8\hat{i} + 2\hat{j} + 6\hat{k}$, find the work done by the force. (DUHS 2009)
 A 50 units C 125 units
 B 75 units D 155 units
 E 200 units

15. Quantities which can be specified by a number having appropriate units (positive, negative, zero) are called:
 A Scalars B Vectors C Physical units D Chemical units
16. The addition, subtraction and multiplication of scalars takes place by usual:
 A Theoretical method C Graphical method
 B Arithmetic rules D None of these
17. Physical quantities having both magnitude and direction with appropriate unit are called:
 A Scalars B Vectors C Physical units D Chemical units
18. Generally, a vector is specified by using the
 A Reference axis C Three dimensional axis
 B Direction indicator D Cartesian co-ordinate system
19. A vector in space has:
 A One component C Three component
 B Two component D No component
20. A vector whose magnitude is one is known as:
 A A free vector C Position vector
 B A unit vector D A null vector
21. If a vector is divided by its magnitude, we get:
 A A unit vector C Position vector
 B A free vector D A null vector
22. A vector which can be displaced parallel to itself and applied at any point, is known as:
 A A null vector C A unit vector
 B A free vector D Position vector
23. A vector having zero magnitude is called:
 A Position vector C A null vector
 B A free vector D A unit vector
24. Negative vector $-A$ as compared with vector A has:
 A Same magnitude C Same direction
 B Opposite direction D Both A & B
25. If we multiply a vector A by -1 , then the new vector has opposite direction but has:
 A Different magnitude C Greater magnitude
 B Same magnitude D Smaller in magnitude
26. Which of the following is a scalars quantity?
 A Displacement B Electric field C Acceleration D Work
27. Which of the following is a vectors quantity?
 A Speed B Velocity C Torque D Displacement
28. The lines represent north, south, east and west direction or horizontal and vertical axes or x and y axes are:
 A Frame of direction C Line of direction
 B Frame of reference D Reference lines

29. In head to tail rule the line which join tail of the first vector with the head of the last vector is called:
 A Reference line B Final direction C Resultant vector D Normal line
30. The sum of two or more vectors is a single vector which is called:
 A A position vector C A negative vector
 B A resultant vector D A null vector
31. If the resultant of two vectors each of magnitude F is also of magnitude F, the angle b/w them will be:
 A 90° B 60° C 30° D 120°
32. The resultant of two forces of 3 N and 4 N making an angle 90° with each other is:
 A 1 N B 7 N C 5 N D 10 N
33. If the resultant of two vectors each of magnitude F, is twice of magnitude F, the Angle b/w them will:
 A 0° B 60° C 90° D 120°
34. The angle b/w two rectangular component of any vector in:
 A 30° B 60° C 90° D 120°
35. The vector A makes an angle θ with x-axis then its x component is given by:
 A $A \cos \theta$ B $A \sin \theta$ C $\theta \cos A$ D $\theta \sin A$
36. The vector A makes an angle θ with x-axis then its y component is given by:
 A $A \cos \theta$ B $A \sin \theta$ C $\theta \cos A$ D $\theta \sin A$
37. If $|A + B| = |A - B|$ the angle b/w the vector is:
 A 0° B 60° C 90° D 45°
38. A force of magnitude 30 N making an angle of 60° with x-axis, its:
 A 15 N B 7 N C 5 N D 10 N
39. The vector whose X and Y component are A_x and A_y is given by:
 A $(A_x + A_y)^2$ C $\sqrt{A_x^2 + A_y^2}$
 B $\sqrt{A_x + A_y}$ D $A_x^2 + A_y^2$
40. The direction of the vector A when X and Y component are given:
 A $\theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$ C $\theta = \tan^{-1} \left(\frac{X}{Y} \right)$
 B $\theta = \tan^{-1} \left(\frac{A_x}{A_y} \right)$ D None of these
41. Geometrical method of addition of vector is:
 A Rectangular component method C Multiplication inverse
 B Associative property D Additive inverse
42. Scalar product of two vectors results in a scalar, the product is
 A Vector product B Sine product C scalar product D Cross product
43. Scalar product of two vector A and B is given by
 A $AB \cos \theta$ C $|A| |B| \cos \theta$
 B $A \times B \sin \theta$ D $|A| |B| \sin \theta$

44. The property $A \cdot B = B \cdot A$ in scalar product is called:
 A Commutative property C Multiplication inverse
 B Associative property D Additive inverse
45. Which of the following is correct?
 A $\hat{i} \cdot \hat{j} = \hat{k}$ B $\hat{i} \cdot \hat{j} = 1$ C $\hat{i} \cdot \hat{j} = 0$ D $\hat{i} \cdot \hat{j} = \hat{i}$
46. If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along X, Y, Z axis respectively then their mutual dot product is equal to:
 A 0 B 1 C -1 D 2
47. If $A = A\hat{i}$, $B = B\hat{j}$, then $A \cdot B$ is equal to:
 A Zero B A C A^2 D -A
48. The scalar product of vector $A = 2\hat{i} + 5\hat{k}$ and $B = 3\hat{i} + 4\hat{k}$ is:
 A 26 B 23 C 15 D 20
49. Which of the following vectors is / are perpendicular to the vector $4\hat{i} - 3\hat{j}$?
 A $6\hat{i}$ B $6\hat{j}$ C $4\hat{i} + 3\hat{j}$ D $3\hat{i} + 4\hat{j}$
50. The angle b/w the two vector $-2\hat{i} + 3\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$ is:
 A 0° C 180°
 B 90° D None of the above
51. A body constrained to move in the y-direction, is subjected to a force $F = -2\hat{i} + 15\hat{j} + 6\hat{k}$ N. The work done by this force in moving the body a distance of 10 m along the y-Axis is:
 A 190 J B 160 J C 20 J D 150 J
52. The scalar product of two equal vectors i.e. $A \cdot A$ is equal to:
 A A vector C Square of the vector
 B Unit vector D Null vector
53. The projection of vector $A = \hat{i} - 2\hat{j} + \hat{k}$ onto the direction of $B = 4\hat{i} - 4\hat{j} + 7\hat{k}$ is:
 A $1/9$ B $2/9$ C $9/19$ D $19/9$
54. One of the two rectangular component of a force is 20N and it make an angle of 30° with the force the magnitude of the other component is:
 A 20 B $1\sqrt{3}$ C $20\sqrt{3}$ D $20/\sqrt{3}$
55. When product of vectors results in a vector then the product is called:
 A Vector product C Scalar product
 B Dot product D Vector double product
56. The unit vector parallel to a vector $A = 3\hat{i} + 6\hat{j} - 2\hat{k}$ is:
 A $\hat{i} + \hat{j} + \hat{k}$ C $3/7\hat{i} + 3/7\hat{j} - 4/7\hat{k}$
 B $3/7\hat{i} + 6/7\hat{j} - 2/7\hat{k}$ D $3/49\hat{i} + 6/49\hat{j} - 2/49\hat{k}$
57. If $A \times B = 0$ and $A \neq B$ then the vector are:
 A Parallel B Anti-parallel C Perpendicular D Both A & B
58. $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} =$
 A 0 B 1 C 2 D Constant
59. If $(a \times b)$ points along positive z-axis, then the vectors a and b must lie in:
 A xz-plane B yz-plane C xy-plane D None of them

60. Area of the parallelogram in which the two adjacent sides are A and B is given by:
 A $AB \sin \theta$ B AB C $AB \cos \theta$ D Zero
61. The cross product ($A \times B$) of two non-zero parallel vector is equal to:
 A $AB \sin \theta$ B AB C $AB \cos \theta$ D Zero
62. Which of the following is true?
 A $A \times B = B \times A$ C $A \times B = C \times B$
 B $A \times B = -B \times A$ D $A \times B = C \times A$
63. The cross product of vector F with itself is equal to:
 A F B 2F C Zero D 1
64. A vector of magnitude $5\sqrt{3}$ units and another vector of magnitude 10 units are inclined to each other at angle of 30° . the magnitude of their vector product is:
 A $5\sqrt{3}$ B $25\sqrt{3}$ C 10 D 50
65. A vector A is along the positive x-axis. If B is another vector such that $A \times B$:
 A $4\hat{j}$ B $-4\hat{i}$ C $-(\hat{i} \times \hat{j})$ D $(\hat{j} \times \hat{k})$
66. Two sides of a triangle are formed by the vector $A = 3\hat{i} + 6\hat{j} - 2\hat{k}$ and $B = 4\hat{i} - \hat{j} + 3\hat{k}$, then $A \times B$ will be:
 A $16\hat{i} - 17\hat{j} - 27\hat{k}$ B $16\hat{i} - 17\hat{j} - 6\hat{k}$ C $6\hat{i} - 7\hat{j} - 2\hat{k}$ D $10\hat{i} - 7\hat{j} + 4\hat{k}$
67. Find the torque of a force $F = -3\hat{i} + \hat{j} + 5\hat{k}$ acting at the point $r = 7\hat{i} + 3\hat{j} + \hat{k}$.
 A $4\hat{i} - 38\hat{j} + 16\hat{k}$ B $4\hat{i} + 4\hat{j} + 6\hat{k}$ C $-21\hat{i} + 4\hat{j} + 4\hat{k}$ D $-14\hat{i} + 34\hat{j} - 16\hat{k}$
68. The scalar product of two vector is $2\sqrt{3}$ and the magnitude of their vector product is 2. The angle between them is:
 A 90° B 60° C 30° D 45°
69. If $A = A_1\hat{i} + A_2\hat{j}$ and $B = B_1\hat{i} + B_2\hat{j}$ are non-parallel vectors, then the direction of $A \times B$ is:
 A Along A B Along x-axis C Along y-axis D Along z-axis
70. $\hat{i} \times \hat{j} =$
 A $\hat{k}$ B $\hat{j}$ C $\hat{k}$ D 1
71. $\hat{j} \times \hat{k} =$
 A $\hat{i}$ B $\hat{j}$ C $\hat{k}$ D 1
72. $\hat{k} \times \hat{i} =$
 A $\hat{i}$ B $\hat{j}$ C $\hat{k}$ D 1

MULTIPLE CHOICE QUESTIONS

1. A part is moving along a circular path of radius r , when it makes half a rotation, the ratio between its displacement and distance is:
 A π B 30π C $2/\pi$ D $\pi/2$
2. A car covers the first half of a certain distance with a speed v_1 and the second half with a speed v_2 . Find the average speed during the whole journey? (DUHS 2009)
 A $\frac{v_1 v_2}{2(v_1 + v_2)}$ C $\frac{2v_1 v_2}{v_1 + v_2}$
 B $\frac{2v_1 v_2}{v_1 + v_2}$ D $\frac{v_1 + v_2}{5v_1 v_2}$
 E $\frac{v_1 - v_2}{5v_1 + 9v_2}$
3. A spacecraft exploring a distant planet releases a probe to explore the planet's surface. The probe falls freely a distance of 40 meters during the first 4.0 second after its release. What is the acceleration due to gravity on this planet?
 A 4.0 m/s^2 B 5.0 m/s^2 C 10 m/s^2 D 16 m/s^2
4. A body is thrown vertically upward with initial velocity 9.8 m Sec^{-1} . It will attain height:
 A 9.8 m B 19.6 m C 29.4 m D 4.9 m
5. An object is dropped, the ratio of their heights covered in 1 sec, 2 sec and 3 sec will be:
 A $1:2:3$ B $1:4:6$ C $1:4:9$ D $9:4:1$
6. Car A moves with constant speed of 60 km/h , another car B moves 1 hour later with constant speed of 75 km/h , how much time will take car B, to overtake car A?
 A 5 h B 4 h C 3 h D 2 h
7. You have two balls of identical volume, one weighing 2 Newtons and the other 10 Newtons. Both are falling freely after being released from the same point simultaneously. Which of the following will then be true? (DUHS 2009)
 - I. The 10-N ball falling freely from rest will be accelerated at a greater rate than the 2-N ball.
 - II. At the end of 4 s of free fall, the 10-N ball will have 5 times the momentum of the 2-N ball.
 - III. At the end of 4 s of free fall, the 10-N ball will have the same kinetic energy as the 2-N ball.
 - IV. The 10-N ball processing greater inertia than 2-N ball.
 A I, II and III only C II and IV only
 B I and III only D IV only
 E None of the above
8. A car is traveling on a level highway at a speed of 15 meters per second. A braking force of 3000 Newton brings the car to a stop in 10 seconds. The mass of the car is:
 A 1500 kg B 2000 kg C 2500 kg D 3000 kg



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20. A car going around a certain curve at a speed of 25 mi/hr has a centripetal force acting on it of 100 lb. If the speed of the car is doubled, the centripetal force.
- Is quadrupled
 - Is doubled
 - Is reduced to $1/2$ of the original value
 - Is reduced to $1/4$ of the original value
21. An object whose mass is 100 gm starts from rest and moves with constant acceleration of 20cm/sec^2 . At the end of 8 sec its momentum is, in gm-cm/sec.
- 500
 - 8000
 - 16,000
 - 33,000
22. The shortest distance between two points is known as:
- Velocity
 - Speed
 - Acceleration
 - Displacement
23. The rate of change of displacement is defined as:
- Speed
 - Velocity
 - Acceleration
 - Retardation
24. The distance traveled by a body in one second is called:
- Speed
 - Velocity
 - Average velocity
 - Acceleration
25. The rate of change of velocity is defined as:
- Speed
 - Velocity
 - Acceleration
 - Momentum
26. The rate of decrease in velocity is called:
- Speed
 - Velocity
 - Acceleration
 - Retardation
27. Velocity and acceleration are:
- Both vector quantities
 - Only velocity is vector quantity
 - Only acceleration is vector quantity
 - Both are scalar quantities
28. If the values of instantaneous and average velocities are equal. The body is said to be moving with:
- Uniform acceleration
 - Uniform velocity
 - Uniform speed
 - Variable velocity
29. If a body is moving with constant velocity, then:
- Its direction may be changing
 - Its acceleration is variable
 - Its acceleration is zero
 - None of these
30. An object with a constant speed:
- Is not accelerated
 - Might be accelerated
 - Is always accelerated
 - Also has a constant velocity
31. If a body is moving along a circle with constant speed, then:
- Its acceleration is zero
 - Its acceleration is increasing
 - Its velocity is uniform
 - Its velocity is changing
32. If a body is moving with constant speed in a circle, then its acceleration is directed:
- Towards the centre
 - Away from the centre
 - Tangent to the circle
 - None of these
33. What must be changing when a body is accelerating uniformly?
- The speed of the body
 - The mass of the body
 - The velocity of the body
 - The force acting on the body

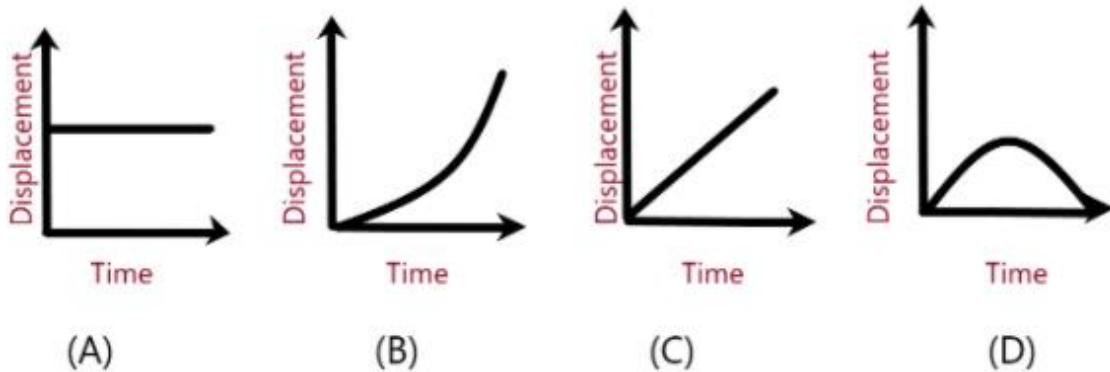
34. If the slope of the velocity-time graph increases at constant rate. With time, then the body is said to have:
- A Uniform retardation C Uniform positive acceleration
B Average acceleration D Uniform negative acceleration
35. Acceleration of Bodies of different masses allowed to fall freely is:
- A Different for different heights C Variable
B Different for different bodies D The same
36. A body of mass 5kg starts from rest and falls freely. The distance covered by the body in one second is:
- A 4.9 m B 49 m C 980 m D 9.8 m
37. A stone is dropped from a cliff. The time during which it covers a distance of 490 m is:
- A 100 sec B 10 sec C 9.8 sec D 4.9 sec
38. A ball is hit so that it moves straight up in air and reaches the maximum height in 2 seconds. Its initial velocity is:
- A 9.8 m/s B 39.2 m/s C 19.6 m/s D 4.9 m/s
39. A body is thrown upward with initial velocity 9.8 m/s. It will go:
- A 19.6 m high B 29.4 m high C 9.8 m high D 4.9 m high
40. When a body is accelerated?
- A Its velocity always changes C Its speed may or may not change
B Its speed always changes D A and C both
41. First Law of Motion is also known as law of:
- A Force B Momentum C Energy D Inertia
42. It is difficult to stop a heavy body as compared to a light body due to:
- A Acceleration B Inertia C Weight D Gravity
43. Mass of a body is a measure of its:
- A Velocity B Acceleration C Inertia D Force
44. The agent which changes or tends to change the state of rest or of uniform motion of a body is called:
- A Inertia C First law of motion
B Momentum D Momentum
45. If a force is applied on a body, an acceleration is produced:
- A In the direction of force C Perpendicular to the direction of force
B Against the direction of force D No acceleration will be produce
46. SI unit of force is:
- A Dyne B Slug C Newton D Newton-meter
47. Mathematical form of Newton's second law is:
- A $a = F/m$ B $a = m/F$ C $F = a/m$ D $a = Fm$
48. Action and reaction are equal and opposite but they act on:
- A Light bodies B Heavy bodies C Different bodies D The same body
49. Newton's second law of motion is also called:
- A Law of Inertia C Law of gravitation
B Law of acceleration D Law of thermodynamics

50. The quantitative measure of inertia of a body is:
 A Its mass B Its weight C Its velocity D Force
51. The laws of motion give the relation between:
 A Mass and weight C Force and acceleration
 B Velocity and acceleration D Mass and velocity
52. In equation $F = ma$, the mass m is:
 A Rest mass C Gravitational mass
 B Inertial mass D Variable mass
53. Laws of motion are valid in a frame of reference which is:
 A At rest B Non-inertial C Inertial D None of these
54. Laws of motion are not valid in a system which is:
 A At rest C Inertial
 B Non-inertial D Moving with uniform
55. The acceleration of a body is zero when a body is:
 A At rest C Both A and B
 B Moving with uniform velocity D None of these
56. Newton's third law of motion states that:
 A Inertia and force are same
 B Inertia and mass are same
 C Action and reaction produce the same acceleration in two bodies
 D When two bodies interact, action and reaction are equal and opposite
57. Which law is applicable in the motion of the rocket in space?
 A Conservation of mass
 B Conservation of energy
 C Conservation of linear momentum
 D Conservation of angular momentum
58. Mathematically Newton's second law in terms of momentum is given by:
 A $F = \Delta p / \Delta v$ C $F = \Delta p / \Delta t$
 B $F = \Delta t / \Delta p$ D None of these
59. Passengers should not stand in a moving bus because:
 A Bus may become unbalanced.
 B Passengers may fall forward as bus suddenly stops
 C Bus may not move fast.
 D None of these
60. Mass is a:
 A Scalar quantity C Variable quantity
 B Vector quantity D Product of two quantities
61. Mass can be measured with the help of:
 A Screw gauge C Spring balance
 B Vernier caliper D Ordinary balance

62. If you compare the mass of a body on the earth and on the moon, then it will be:
 A Greater on earth C Greater on moon
 B Zero on earth D Same on earth and moon
63. Weight of a body is equal to the force with which the earth attracts the body towards it:
 A South Pole B North Pole C Equator D Centre
64. If a body hang vertically with the help of a string, then its weight will act downward and tension in the string will act
 A Rightwards C Upwards
 B Leftwards D In the same direction
65. Dimension of impulse is similar to the dimension of:
 A Momentum B Force C Work D Torque
66. Change in the momentum is called:
 A Impulse B Force C Torque D Energy
67. Rate of change in the momentum is called:
 A Impulse B Force C Torque D Energy
68. A system on which no external agency exerts any force is called:
 A Closed system B Open System C Isolated system D Adiabatic system
69. The total linear momentum of an isolated system remains constant. This is the statement of:
 A Law of conservation of energy
 B Law of conservation of linear momentum
 C Law of conservation of angular momentum
 D Law of conservation of mass
70. A collision in which both K.E. and momentum are conserved is called:
 A Inelastic collision B Elastic Collision C Both of these D None of these
71. When a bullet fired by a gun, the gun recoils backward with a velocity:
 A Less than that of the bullet C Greater than that of the bullet
 B Equal to that of the bullet D None of these
72. The bullet fired from a gun has momentum which is:
 A Equal to recoil of gun C Less than recoil of gun
 B Greater than recoil of gun D Anything is possible
73. In an inelastic collision:
 A Momentum is less after the Collision C Kinetic energy is less after the collision
 B Momentum is more after the collision D Kinetic energy is more after the collision
74. When two bodies are connected with the help of a string which passes over a pulley, then the value of tension in the string is:
 A $T = \frac{2m_2g}{m_1+m_2}$ B $T = \frac{2m_1g}{m_1+m_2}$ C $T = \frac{2m_1m_2g}{m_1-m_2}$ D $T = \frac{2m_1m_2g}{m_1+m_2}$
75. The resultant force for a body which is moving downward when two bodies are connected with a string which passes over a pulley is given by the expression.
 A $mg + ma = -T$ C $mg - T = ma$
 B $mg - ma = -T$ D $mg - T = -ma$

76. When two bodies connected with an inextensible string which passes over a pulley and one body moves vertically and other horizontally on the table, then the value of tension in the string is:
- A $T = \frac{2m_2g}{m_1+m_2}$ B $T = \frac{m_1m_2g}{m_1+m_2}$ C $T = \frac{2m_1m_2g}{m_1-m_2}$ D $T = \frac{2m_1m_2g}{m_1+m_2}$
77. When one body moves vertically downward and other upward which are connected with a string passes over the pulley, the expression for acceleration is:
- A $a = \left(\frac{m_1+m_2}{m_1-m_2}\right)g$ B $a = \left(\frac{2m_1m_2}{m_1+m_2}\right)g$ C $a = \left(\frac{m_1}{m_1+m_2}\right)g$ D $a = \left(\frac{m_1-m_2}{m_1+m_2}\right)g$
78. When one body moves vertically downward and other moves on a smooth horizontal plane, which are connected with a string passes over the pulley, the expression for acceleration is
- A $a = \left(\frac{m_1+m_2}{m_1-m_2}\right)g$ B $a = \left(\frac{2m_1m_2}{m_1+m_2}\right)g$ C $a = \left(\frac{m_1}{m_1+m_2}\right)g$ D $a = \left(\frac{m_1-m_2}{m_1+m_2}\right)g$
79. Which of the following is always equal to magnitude of force but opposite in direction?
- A Tension B Velocity C Acceleration D Mass
80. The force which resists the motion of one surface on another surface is called:
- A Normal force B Force of friction. C Resultant force D Applied force
81. The relation between kinetic and static friction is:
- A $f_k = f_s$ B $f_k > f_s$ C $f_k = f_s = 0$ D $f_k < f_s$
82. When we try to move a body from rest then a force comes into play and pulls the body in the opposite direction, this force is the:
- A Friction B Static friction C Kinetic friction D Normal friction
83. When we apply a force on a body to move it from rest, initially it will not move but at certain value of applied force body starts moving, this maximum value of static friction is known as:
- A Limiting friction B Maximum friction C Minimum friction D Zero friction
84. When a body is moving over the other, the force of friction between them is known as:
- A Static friction B Maximum friction C Kinetic friction D Limiting friction
85. Stoke's law is applicable to:
- A Motion through non-resistive medium C Motion through free space
B Motion through viscous medium D Motion through horizontal surface
86. The drag force F_D on a sphere of radius r moving with velocity v as given by Stoke's law is:
- A $F_D = 6\pi\eta r^2v$ B $F_D = 6\pi\eta^2rv^2$ C $F_D = 6\pi/\eta rv$ D $F_D = 6\pi\eta rv$
87. A car covers the first half of a certain distance with a speed v_1 and the second half with a speed v_2 . The average speed during the whole journey is
- A $\frac{(v_1+v_2)}{2}$ B $\frac{(v_1-v_2)}{2}$ C $\frac{2v_1v_2}{(v_1+v_2)}$ D $\frac{v_1+v_2}{v_1v_2}$
88. A vehicle starting from rest attains a speed of 72 km/h after covering a distance of 100m. If the mass of vehicle is 500kg. The force exerted by the engine is
- A 100 N B 500 N C 1000 N D 2000 N
89. A mass of 75 kg stands in a lift. What force does the floor exert on him when the lift starts moving upward with an acceleration of 2 m/s^2 ($g = 10 \text{ m/s}^2$)
- A 1200 N B 900N C 1600 N D 2000 N

90. A box of mass $m = 6\text{ kg}$ slides with speed $v = 4\text{ m/s}$ across a frictionless floor. If suddenly explodes into two pieces. One piece, with mass $m_1 = 2\text{ kg}$ moves in the same direction with speed $v_1 = 8\text{ m/s}$, the velocity of the second piece is:
 A 1 m/sec B 2 m/sec C 3 m/sec D 4 m/sec
91. The speed-time graph of a particle moving along a straight line is shown below. The distance covered by the particle in 10 s is:
 A 5 m B 50 m C 100 m D 150 m
92. Which of the following graphs represents the displacement (s)- time (t) graph for uniform motion?



93. The displacement-time graph for two particles A and B are straight lines inclined at 60° and 30° to the time axis. The ratio of their speed is:
 A 3:1 B 4:1 C 5:1 D 6:1

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MULTIPLE CHOICE QUESTIONS

1. A ball "P" is dropped vertically from the top of a building 49m high, at the same time another ball 'Q' is thrown horizontally with a speed of 21m/s (Neglecting the resistance of air)
 - A Ball 'P' will reach first than ball 'Q'
 - B Ball 'Q' will reach first than ball 'P'
 - C Both balls will reach the ground at the same time
 - D None of these
2. A bomb is dropped from an airplane moving horizontally with a speed of 600 km/h. If the air resistance is negligible, the bomb will reach the ground in 5s when the attitude of the plane is approximately
 - A 50 m
 - B 75 m
 - C 125 m
 - D 250 m
3. A rescue helicopter drops a package of emergency ration to a standard party on the ground. If the helicopter is traveling horizontally at 40m/s at a height 100 m above the ground, where does the package strike the ground relative to the point at which it was released? ($g = 9.8 \text{ m/s}^2$) (DUHS 2009)
 - A 120 m
 - B 130 m
 - C 140 m
 - D 180.7 m
 - E 200.3 m
4. A ball is thrown with a horizontal velocity of 6.0 m/s. What is horizontal velocity after 3.0 second of flight? (Take $g = 10 \text{ m/s}^2$)
 - A 36 m/s
 - B 15.8 m/s
 - C 6 m/s
 - D 4.9 m/s
5. A shot leaves a gun at the rate of 160 m/s. Calculate the greatest distance to which it could be projected and the height to which it would rise? ($g = 10 \text{ m/s}^2$) (DUHS 2009)
 - A 1560 m, 540 m
 - B 2560 m, 640 m
 - C 3560 m, 740 m
 - D 4560 m, 840 m
 - E 9595 m, 348 m
6. At what range of Projectile horizontal distance equals to vertical distance:
 - A $\tan^{-1}(0)$
 - B $\tan^{-1}(1)$
 - C $\tan^{-1}(2)$
 - D $\tan^{-1}(4)$
7. The horizontal range of a shell fired with a speed of 49 ms^{-1} at an angle of 45° with the horizontal is:
 - A 49 m
 - B 490 m
 - C 245 m
 - D 2401 m
8. A projectile is fired at $\pi/3$ radians with range R_1 and another projectile is fired at $\pi/6$ radians with range R_2 , then the relation is:
 - A $R_1 = R_2$
 - B $R_1 = 2R_2$
 - C $R_1 = 4R_2$
 - D $R_1 = 3R_2$
9. An object travels at constant speed around a circle of radius 1.0 m in 0.1s. What is the magnitude of its acceleration? (LUHMS 2007)
 - A Zero
 - B 1.0 ms^{-2}
 - C $4\pi \text{ ms}^{-2}$
 - D $4\pi^2 \text{ ms}^{-2}$
10. An object of mass of 2kg rotates at constant speed in a horizontal circle of radius 5m. The time for one complete revolution is 3 s. What is the magnitude of centripetal force action on the object? (LUHMS 2007)
 - A $4\pi^2/9 \text{ N}$
 - B $40\pi^2/9 \text{ N}$
 - C $100\pi^2/9 \text{ N}$
 - D $400\pi^2/9 \text{ N}$

11. If a car moves with a uniform speed of 2m/sec in a circle of radius 0.4m. Its angular speed is:
 A 4 rad/sec B 5 rad/sec C 2.5 rad/sec D 6 rad/sec
12. A cyclist moves in a circular track of radius 50 meters. If the coefficient of friction is 0.2, then the maximum velocity with which the cyclist can take the turn without slipping is:
 A 10 m/s B 5 m/s C 20 m/s D 7.5 m/s
13. A cyclist turn around a curve at 20 km/hr, if he turns his double the speed, the tendency to overturn will be:
 A Double B Quadruple C the same D Halved
14. A point of the rim of a wheel moves 2.5 m when the wheel turns through an angle of 0.1 rad. What is the radius of the wheel?
 A 0.5 m B 2 m C 0.2 m D 20 m
15. A wheel is a 1 m in diameter, when it makes 30 rev. per mint, the liner speed of a point in m/sec on its circumference is:
 A π B 30π C $\pi/2$ D 60π
16. What is the angular velocity of second hand watch?
 A $\pi/\text{rad/s}$ B $\pi/30\text{rad/s}$ C $\pi/2 \text{ rad/s}$ D $60\pi \text{ rad/s}$
17. In projectile motion the component which remain constant is:
 A Horizontal velocity component C Both A and B
 B Vertical velocity component D None of these
18. During the projectile motion, the horizontal component of velocity
 A Change with Time C Become zero
 B Remains Constant D Decreases with time
19. The velocity of a projectile is maximum:
 A At half of the height
 B At one fourth of the height
 C Just before striking the ground and point of Projectile
 D None of these
20. The horizontal range of a projectile, at a certain place, depends upon:
 A The mass of the projectile
 B The velocity of projection
 C The angle of projection
 D The angle as well as velocity of projection
21. The vertical component of velocity of a projectile, during its motion, is maximum:
 A At the highest point of its trajectory.
 B At the point of projection
 C Just before striking the ground and point of projection
 D None of these
22. At maximum height the velocity component which becomes zero is:
 A Horizontal velocity component C Both A and B
 B Vertical velocity component D None of these
23. An object thrown in arbitrary motion, in space with an initial velocity and moving freely under gravity will follow:
 A A circle B A hyperbola C A parabola D A straight line

24. A man in a train moving with a constant velocity drops a ball on the platform. The path of the ball as seen by an observer standing on the platform is the two angles of Projection for which the range is 20m ? ($g = 10 \text{ m/sec}^2$)
- A A circle
B A straight line
C A parabola
D None of these
25. The velocity of Projectile is maximum at:
- A Point of launch
B Highest point
C Point of landing
D Both A and C
26. The projectile attains maximum horizontal range when it is projected at an angle of:
- A 90° with the horizontal
B 60° with the horizontal
C 45° with the horizontal
D 30° with the horizontal
27. To improve the jumping record, a long jumper should:
- A Jump at an angle of 45°
B Jump at an angle of 60°
C Jump as high as possible
D Jump at an angle of 30°
28. The horizontal ranges is equal for the angles:
- A 0° and 90°
B 45° and 90°
C 30° and 45°
D 30° and 60°
29. A Ball is projected with a speed of 10 m/s . The angles of Projection for which the range is 5.0 m are ($g = 10 \text{ m/s}^2$).
- A $30^\circ, 60^\circ$
B $15^\circ, 75^\circ$
C $45^\circ, 45^\circ$
D $10^\circ, 80^\circ$
30. Two Projectile are fired from the same point with the same speed at angles of Projection 60° and 30° respectively. Which one of the following is true?
- A Their Time of flight will be same
B Their landing velocity will be same
C Their range will be same
D Their maximum height will be same
31. The bomber drops a bomb when it is vertically above the target, it missed:
- A Velocity may vary
B Acceleration changes
C Vertical velocity changes
D Horizontal velocity changes
32. The maximum height attained by the projectile is given by:
- A $H = v_o^2 \cos^2 \theta / g$
B $H = v_o^2 \sin^2 \theta / g$
C $H = v_o^2 \sin^2 \theta / 2g$
D $H = v_o^2 \cos^2 \theta / 2g$
33. The time taken to achieve maximum height is given by:
- A $t = v_o \cos \theta / g$
B $t = v_o \sin \theta / 2g$
C $t = v_o \cos \theta / 2g$
D $t = v_o \sin \theta / g$
34. The total time of flight of Projectile is given by:
- A $T = v_o \sin \theta / 2g$
B $T = v_o \cos \theta / g$
C $T = 2v_o \cos \theta / g$
D $T = 2v_o \sin \theta / g$
35. The horizontal distance between the point of projection i.e. range is given by:
- A $R = v_o^2 \sin 2\theta / g$
B $R = v_o^2 \cos 2\theta / g$
C $R = v_o^2 \sin \theta / g$
D $R = v_o^2 \cos \theta / g$
36. The maximum range of Projectile is given by:
- A $R_{\max} = v_o^2 / g$
B $R_{\max} = v_o^2 \sin \theta / g$
C $R_{\max} = v_o \sin \theta / g$
D $R_{\max} = v_o^2 \cos \theta / g$

37. For a projectile, the ratio of maximum height reached to the square of flight time is ($g = 10 \text{ m/s}^2$):
 A 5:2 B 5:4 C 5:1 D 10:1
38. An object is thrown along a direction inclined at an angle of 45° with the horizontal. The horizontal range of the object is equal to:
 A Twice the vertical height C Vertical height
 B Four times the vertical height D Thrice the vertical height
39. The horizontal and vertical components of the initial velocity of a projectile are 10 m/s and 20 m/s , respectively. The horizontal range of the projectile will be ($g = 10 \text{ m/s}^2$):
 A 5 m B 10 m C 20 m D 40 m
40. A ball is projected with a speed of 10 m/s . The two angles of Projection for which the range is 5 m are ($g = 10 \text{ m/s}^2$):
 A $15^\circ, 75^\circ$ B $30^\circ, 60^\circ$ C $45^\circ, 45^\circ$ D $10^\circ, 80^\circ$
41. A ball is projected at an angle of 45° to the horizontal. If the range is 10 m . The maximum height attained by the ball is ($g = 10 \text{ m/s}^2$)
 A 1 m B 2 m C 2.5 m D 3 m
42. The angle of projection for which the maximum height and the horizontal range of a projectile are equal is:
 A 45° B $\tan^{-1}(4)$ C $\tan^{-1}(1/4)$ D None of these
43. If the horizontal range of Projectile is four times its maximum height, the angle of projection is:
 A 30° B $\tan^{-1}(1/4)$ C $\sin^{-1}(1/4)$ D 45°
44. The angle of projection of a projectile is 30° . The ratio of the horizontal range and the maximum height is:
 A $4\sqrt{3}$ B $3\sqrt{4}$ C $2\sqrt{3}$ D $4\sqrt{2}$
45. The angle subtended at the centre by a body moving along a circle is called:
 A Angular displacement C Angular speed
 B Angular Acceleration D Angular Velocity
46. The rate of change of angular displacement is called:
 A Angular displacement C Angular speed
 B Angular Acceleration D Angular Velocity
47. The S.I unit of angular displacement is:
 A Centimeter B Radian C Kelvin D Meter
48. The angle subtended at the centre of a circle by an arc of Length equal to its radius is equal to:
 A One rotation B One degree C One radian D Half rotation
49. The relation between time T and angular velocity ω is given by:
 A $T = \omega / 2\pi$ B $T = 2\pi / \omega$ C $T = 2\pi\omega$ D $T = v\omega$
50. The direction of linear velocity of a body at a point moving along a circular path is:
 A Along the axis of rotation C Directed away from the centre
 B Directed towards the centre D Along the tangent

51. When a body moves in a circle, the angle between its linear velocity v and the angular velocity ω is:
 A 90° B 0° C 45° D 180°
52. When an object moves with uniform speed in a circular orbit, its centripetal acceleration must be directed:
 A Away from the centre of circle C Towards the centre of circle
 B Perpendicular to the centre of circle D Along the direction of motion
53. The force needed to maintain the circular motion of a body is called:
 A Centripetal force C Coulomb force
 B Centrifugal Force D Gravitational force
54. The acceleration of a body undergoing uniform circular motion is constant in:
 A Both direction and magnitude C Direction only
 B Neither direction nor magnitude D Magnitude only
55. When a stone is whirled in a horizontal circle by means of a string, the centripetal force is supplied by:
 A Tension in the string C Velocity of the stone
 B Mass of the stone D Air friction
56. A stone is tied at the end of a cord and is whirled from the other end in a circle at constant speed. If the cord breaks, the stone moves along:
 A The radius away from the centre C The tangent
 B The radius towards from the centre D The direction of motion
57. When a stone is whirled in a vertical circle at the end of a string, the tension in the string is maximum:
 A At the top C At the horizontal diameter
 B At the bottom D At the centre
58. Mathematically angular Displacement is defined as:
 A $S = \text{radius} \times \theta$ (in radians) C $S = \text{circumference} \times \theta$ (in radians)
 B $S = \text{radius} \times \theta$ (in degree) D $S = \text{circumference} \times \theta$ (in degree)
59. A car moving with a uniform speed of 2 m/s in a circle of radius 0.4 m. Its angular speed in rad/sec is:
 A 1.6 B 4 C 0.8 D 5
60. A point on the rim of a wheel moves 0.2 m when the wheel turns through an angle of 0.1 rad. What is the radius of the wheel?
 A 0.5 m B 2 m C 0.2 m D 20 m
61. A car of mass 1 kg moves round a curve whose radius of curvature is 100 m at a speed 10 m s^{-1} . What is the centripetal force on the car?
 A 1 N B 0.1 N C 10 N D 100 N
62. A particle revolves round a circular path with a constant speed. The acceleration of the particle is:
 A Zero C Along the circumference of the circle
 B Along tangent D Along the radius

63. If the body moves with a constant speed in a circle:
- A No work is done on it
 - B No force acts on it
 - C No acceleration is produced in it
 - D Its velocity remains constant
64. For a particle moving in uniform circular motion:
- A Velocity is transverse and acceleration is radial
 - B Velocity is radial and acceleration is transverse
 - C Both velocity and acceleration are transverse
 - D Both velocity and acceleration are radial
65. A ball is thrown horizontally from the top of a cliff with a speed of 40m/s. If it hits the ground in 2 seconds. What is the distance from the cliff to the point at which the ball hits the ground?
- A 19.6 m
 - B 20 m
 - C 40 m
 - D 80 m



MULTIPLE CHOICE QUESTIONS

1. A body is called a rigid body if under the action of a force distance between two points.

A Increases	C Changes
B Decreases	D Remains the same
2. A force passing through the centre of gravity of a body:

A Holds the body in equilibrium	B Results only in rotational motion
C Results only in translational motion	D Results in both translational and rotational motion
3. The straight line joining all the centre of circles in which particles of a rigid body move during rotation is:

A Axis of symmetry	B Axis of rotation	C Parallel line	D Moment arm
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4. If the axis of rotation lies outside the body the corresponding rotatory motion is called:

A Orbital motion	C Vibratory motion
B Spin Motion	D To-and-fro motion
5. If the axis of rotation passes through the body itself the corresponding rotatory motion is called the:

A Orbital motion	C Vibratory motion
B Spin Motion	D To-and-fro motion
6. Torque is also called as:

A Moment arm	C Moment of force
B Moment of inertia	D Angular Velocity
7. Torque produced by a force is equal to the product of:

A Magnitude of force and moment arm	B Magnitude of force and angular velocity
C Magnitude of force and acceleration	D Magnitude of force and displacement
8. Torque is given by:

A $\tau = F \times r$	B $\tau = F \times r$	C $\tau = r \cdot F$	D $\tau = F \cdot r$
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9. Torque produced by force depends upon:

A Speed of body	B Force applied	C Moment arm	D Both B and C
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10. The perpendicular distance from axis of rotation and line of action of force is:

A Moment of force	C Moment arm
B Displacement	D Perpendicular distance
11. If the line of action of force passes through the axis of rotation then moment arm is:

A One	B Zero	C Minimum	D Maximum
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12. If the line of action of force passes through the axis of rotation then torque is:

A One	B Zero	C Minimum	D Maximum
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13. Torque is zero when angle between r and F is:

A 0°	B 45°	C 60°	D 90°
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14. Torque is maximum when angle between r and F is:
 A 0° B 45° C 60° D 90°
15. Anti-clockwise torque is taken as:
 A Zero B Constant C Positive D Negative
16. Clockwise torque is taken as:
 A Zero B Constant C Positive D Negative
17. Force F is applied on body which can rotate about origin then torque is:
 A Rf B $rF \sin \theta$ C $rF \cos \theta$ D None of these
18. The vector sum of two vectors in a couple is:
 A Zero B Anyone of the vector C Magnitude of both D Resultant of vectors
19. A couple produces:
 A No motion B Purely rotational motion C Purely linear motion D Both linear and rotational motion
20. When the external torque on a system is zero there will be conservation of its:
 A Total Energy B Linear Momentum C Angular Momentum D None of these
21. The centre of gravity of a body is:
 A The point at which the whole weight of a body acts
 B The centre of the body
 C The point at which the mass of body acts
 D None of these
22. The centre of mass of a body is at:
 A The middle of the body
 B The point at which the whole mass of a body acts
 C The point at which the whole weight of a body acts
 D The centre of the body
23. Centre of mass of a body can exist:
 A Only outside B Inside C Inside or outside D Neither inside nor outside
24. The knowledge of centre of gravity helps a lot in solving the problems on:
 A Rigid bodies B Optics C Mechanics D Equilibrium
25. The force of gravity acting on a body is equal to its
 A Equilibrium B Centre of gravity C Momentum D Weight
26. The centre of mass of a uniform rod is
 A Mid-point of axis B At its end C Centre of rod D None of these
27. The centre of gravity of a circle lies at
 A Circumference B Its centre C Its radius D Its Diameter
28. The centre of gravity of a cylinder lies at its:
 A Centre B Diameter C Central point of axis D Axis

29. The centre of gravity of a triangular plate lies at the point of:
- A Intersection of two sides C Intersection of medians
B Centre of perpendicular side D Centre of base
30. The centre of gravity of square, rectangle and parallelogram etc. lies at the point of:
- A Intersection of two sides C Intersection of diagonals
B Intersection of one side and its normal D None of above
31. Then centre of mass of a sphere is:
- A At its surface B Centre of sphere C Mid-point of axis D None of these
32. If a number of force are acting on a body and the body is at rest then the body is in state of:
- A Static equilibrium C Dynamic equilibrium
B Stable equilibrium D Unstable equilibrium
33. A book lying on a table is an example of:
- A Static equilibrium C Dynamic equilibrium
B Stable equilibrium D Unstable equilibrium
34. If a number of forces are acting in a body and the body is moving with uniform velocity, then the body is in the state of:
- A Static equilibrium C Unstable equilibrium
B Dynamic equilibrium D Stable equilibrium
35. Motion of the paratrooper is an example of:
- A Static equilibrium C Unstable equilibrium
B Dynamic equilibrium D Stable equilibrium
36. The first condition of equilibrium is:
- A $\Sigma F_x = 0$ B $\Sigma F_y = 0$ C $\Sigma F_x = \Sigma F_y = 0$ D $\Sigma \tau = 0$
37. The second condition of equilibrium is:
- A $\Sigma F_x = 0$ B $\Sigma F_y = 0$ C $\Sigma F_x = \Sigma F_y = 0$ D $\Sigma \tau = 0$
38. When first condition of equilibrium is satisfied then the body has no:
- A Linear acceleration C Rotational acceleration
B Linear velocity D Motion
39. When second condition of equilibrium is satisfied then the body has no:
- A Linear acceleration C Rotational acceleration
B Linear velocity D Motion
40. The body will be in complete equilibrium if:
- A First condition is satisfied only C Both conditions will be satisfied
B Second condition is satisfied only D None of above condition would satisfy
41. The angular momentum is defined as:
- A The product of displacement and linear momentum
B The dot product of positive vector and linear momentum
C The cross product of positive vector and linear momentum
D None of these
42. Torque in terms of angular momentum is given by:
- A $\tau = \Delta L / \Delta t$ B $\tau = \Delta t / \Delta L$ C $\tau = \Delta L \cdot \Delta t$ D $\tau = \Delta L / \Delta t$

43. A constant torque acting on a uniform circular wheel changes its angular momentum from A to $5A$ in three seconds. The magnitude of this Torque is:
 A A B $4A$ C $12A$ D $4A/3$
44. The S.I unit of angular momentum is:
 A J B $J \times \text{Sec}$ C Nm D N/m



MULTIPLE CHOICE QUESTIONS

1. An object weighs 100 Newton on Earth's surface, when it is moved to a point one Earth radius above Earth's surface, it will weigh will be:
 A 100 N B 25 N C 50 N D 400 N
2. Moon has masses $1/96$ of that of the earth and has $1/4$ diameter to that of the earth, the weight of a man on moon will be:
 A Same as that on the surface of Earth C $1/4$ of that on the surface of Earth
 B $1/5$ of that on the surface of Earth D $1/6$ of that on the surface of Earth
3. At what depth from the surface of earth, the value of gravitational acceleration remains one half:
 A $1/2 R_e$ B $1/2 R_e$ C $1/4 R_e$ D $2/3 R_e$
4. A man standing on a scale in an elevator notices that the scale reads 30 Newton greater than his normal weight. Which type of movement of elevator could cause this greater than normal reading.
 A Accelerating downward C Moving upward at constant speed
 B Accelerating upward D Moving downward at constant speed
5. A man weights 980 Newton in a stationary lift. What will be his weight if the lift starts moving up with an acceleration of 9.8 m/sec^2 .
 A 980 N B 1950 N C 1960 N D 1176 N
6. The satellites are in a circular orbit around the earth. What is the relationship between the radii r of their orbits and their speeds v ? (LUMHS 2007)
 A $v \propto r^2$ B $v \propto r$ C $v^2 \propto 1/r$ D $v \propto 1/r^2$
7. Newton's law of gravitation is stated as everybody attracts every other body with a force which is directly proportional to the product of masses and inversely proportional to:
 A Cube of the distance between them.
 B Distance between them
 C Square of product of their radii.
 D Square of the distance between their center
8. Which of the following basic factors in nature is the weakest?
 A Nuclear Force C Gravitational force
 B Electromagnetic force D Electromotive force
9. Mathematically Newton's law of gravitation is given by:
 A $F = G (m_1 + m_2) r^2$ C $F = G (m_1 m_2)/r$
 B $F = G (m_1 m_2)/r^2$ D $F = Gr^2/m_1 m_2$
10. The value of constant of Gravitation is:
 A $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$ C $6.67 \times 10^{11} \text{ Nm}^2 \text{ kg}^{-2}$
 B $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^2$ D $6.67 \times 10^{-11} \text{ Nm kg}^{-2}$
11. Which basic force is responsible for the motion of falling objects and motion of planets around the sun?
 A Strong nuclear force C Gravitational force
 B Weaker nuclear force D Electromotive force

12. The value of gravitational constant G was first determined in the laboratories by:
 A Einstein B Cavendish C Maxwell D Newton
13. The gravitational force between two particles, each of mass 1 kg, separated by a distance of 1m in vacuum is:
 A $6.67 \times 10^{-5} \text{ N}$ B $6.67 \times 10^{-7} \text{ N}$ C Zero D $6.67 \times 10^{-11} \text{ N}$
14. If the distance between two mass particles is halved, the gravitational force between them becomes:
 A Half B Double C Three times D Four times
15. If the distance between two mass particles is doubled and their masses are also doubled, the gravitational force:
 A Is halved C Remains Unchanged (Constant)
 B Is Doubled D Becomes Four times
16. If the distance between two mass particles is doubled, the gravitational force between them:
 A Is reduced to one fourth of the original value
 B Becomes Four times
 C Is reduced to half of the original value
 D Becomes double
17. Which one of the following can be used to determine the mass of the earth?
 A Newton's law of gravitation C Kepler's law
 B Newton's law of motion D Coulomb's law
18. The radius of earth is:
 A $6.64 \times 10^8 \text{ m}$ B $6.64 \times 10^6 \text{ m}$ C $4.6 \times 10^7 \text{ m}$ D $6 \times 10^3 \text{ m}$
19. The mass of earth is:
 A $6 \times 10^{30} \text{ kg}$ B $6 \times 10^{34} \text{ kg}$ C $6 \times 10^{24} \text{ kg}$ D $6 \times 10^{21} \text{ kg}$
20. The mass of Sun is:
 A $3.14 \times 10^{21} \text{ kg}$ B $2.43 \times 10^{27} \text{ kg}$ C $1 \times 10^{30} \text{ kg}$ D $1.99 \times 10^{30} \text{ kg}$
21. The mass of earth can be determined by the equation:
 A $M_e = R_e g / G$ B $M_e = R_e^2 g / G$ C $M_e = R_e^2 G / g$ D $M_e = R_e G / g$
22. The centripetal acceleration of the moon is:
 A $1.2 \times 10^{-3} \text{ ms}^{-2}$ B $1.2 \times 10^{-5} \text{ ms}^{-2}$ C $2.72 \times 10^{-3} \text{ ms}^{-2}$ D $8.5 \times 10^{-6} \text{ ms}^{-2}$
23. Which of the following relation is correct?
 A $a_m / g = 1 / (60)^2$ B $a_m / g = 1 / (80)^2$ C $a_m / g = 1 / (10)^2$ D $a_m / g = 1 / (50)^2$
24. The value of g at the centre of earth is:
 A Zero B 9.8 m/s^2 C 4.9 m/s^2 D 19.6 m/s^2
25. A body weight 100 N on the surface of the earth. It's weight at a depth equal to half the radius of the earth is:
 A 100 N B 25 N C 75 N D 50 N
26. If the mass and diameter of the earth become two times their present values, then the weight of an object on the surface of the earth will:
 A Remain Unchanged C Become half
 B Become one-fourth D Be Doubled



- B By causing the satellite to rotate about its own axis
 C Decreasing the orbital speed of the satellite
 D None of these
36. A body in satellite orbiting round the earth is weightless because
- A The moon attracts the body with a force equal to the force of attraction of the earth
 B The body is free from the force of attraction of the earth
 C The body is free from the force of attraction of the moon
 D The body is free from the force of attraction of the sun

MULTIPLE CHOICE QUESTIONS

1. A body moves a distance of 10m along a straight line under the action of 5N. If the work done is 25 joules, the angle which the force makes with the direction of motion of the body is:
A 0° B 30° C 60° D 90°
2. A body of mass 5 kg, initially at rest is moved by a horizontal force of 2 N on a smooth horizontal surface. Find the work done by the force is 10 s. (DUHS 2009)
A 40J C 50J
B 30J D 20J
E 10J
3. A force of 10 N is required to move an object at a constant speed of 5 m/s. The power used is:
A 5 watts B 2 watts C 16 watts D 50 watts
4. If the K.E of an object is 16 joules when its speed is 4 metre per second then the mass of the object is:
A 1.00 kg B 2 kg C 4.9 kg D 19.6 kg
5. Two bodies of mass 1kg and 2kg have equal Momentum the ratio of their Kinetic energies is:
A 1:1 B 2:1 C 4:1 D 1:4
6. A boy is swinging like a pendulum. He is 2 metre above the ground at mean position and 7 at extreme position. What will be his maximum velocity?
A 100 m/s B 10 m/s C 5m/s D 9m/s
7. A ball is moving horizontally with speed 'v' strikes the bob of a simple pendulum at rest. The mass of the bob is equal to that of the ball. If the collision is elastic the bob will rise to a height. (DUHS 2009)
A v^2/g B $v^2/2g$ C $v^2/4g$ D $v^2/8g$
8. The scalar product of force and displacement is called:
A Power B Work C Acceleration D Momentum
9. Work done under a force is negative if the force and displacement are:
A At the right angle to each other C In opposite direction
B In the same direction D None of these
10. Work done will be positive (maximum) if the angle between force and displacement is:
A 0° B 45° C 90° D 180°
11. Work done against friction is an example of:
A Positive work C Zero work
B Negative work D None of these
12. If the force and displacement are at right angle to each other, then the work done is:
A Zero B Maximum C Negative D Positive
13. Joule is the S.I unit of which pair:
A Force and energy C Work and Energy
B Power and energy D Force and momentum
14. The work done in moving a body along a closed path in a conservative field is:
A Mgh B fh C gh D Zero

15. Work done is said to be one joule when one newton of force acting on a body displaces it through a distance:
- A One meter in direction of force C 1 cm in direction of force
B 1 m in direction opposite of force D 1 cm in the direction opposite of force
16. A body moves at a distance of 10 M along the horizontal by a force of 5 Newton. if the work done is 25 joule then the angle between them is:
- A 0° B 30° C 60° D 90°
17. A body of mass 5 kg initially at rest, is moved by a force of 2 Newton on a smooth horizontal surface. The work done by the force in 10 second is:
- A 20J B 30J C 40J D 60J
18. Work is always done on a body when:
- A A force acts on it
B it moves through a distance
C Its momentum changes
D It experiences and increases in energy through a mechanical influence
19. A body is displayed from $x = x_1$ to $x = x_2$ by a force of $2x$. The work done is:
- A $2x(x_2 - x_1)$ B $2x_2(x_2 - x_1)$ C $x_2^2 - x_1^2$ D $(x_2 - x_1)^2$
20. A body of mass 10 kg is dropped to the ground from a height of 10 m. The work done by the gravitational force is ($g = 9.8 \text{ m/s}^2$)
- A -490J B +490J C -980J D +980J
21. A body is lifted by a man to a height of 1 m in 30s. Another man lift the same as to the same height in 60s. The work done by them are in the:
- A 1:2 B 1:1 C 2:1 D 4:1
22. A force $F = 5\hat{i} + 6\hat{j} - 4\hat{k}$ is acting on a body, produces a displacement $S = 6\hat{i} + 5\hat{j}$. Work done by the force is:
- A 10 units B 18 units C 11 units D 15 units
23. The scalar product of force and velocity is:
- A Power B Energy C Work D Impulses
24. Power is defined as:
- A Capacity to do work C Rate of doing work
B Ability to do work D Habit of doing work
25. The work is done at the ratio of 1000 joules per second, then the power of machine is:
- A One Watt B One kilo-watt C Joule D Horse power
26. The unit of power in British engineering system is:
- A Watt B Kilo-watt C Joule D Horse power
27. One holds power is equal to:
- A 476 watt B 1000 watt C 746 watt D 1 watt
28. And electric motor creates tension of 45 N in a hoisting cable and reels it in at the rate of 2 m/s. The power motor is:
- A 15 KW B 90 watt C 225 watt D 9000 HP

29. The time taken by an engine of power 10kW to lift a mass of 200 kg to a height of 40 m is ($g = 10\text{m/s}^2$):
 A 2s B 4s C 8s D 16s
30. A particle moves with the velocity $6\hat{i} - 4\hat{j} + 3\hat{k}$ m/s under the influence of a constant force $F = 20\hat{i} + 15\hat{j} - 5\hat{k}$ N. The instantaneous power applied to the particle is:
 A 45 J/s B 35 J/s C 25 J/s D 195 J/s
31. The kinetic energy of the body depends upon:
 A Force and acceleration of the body C Mass and velocity of the body
 B Density and weight of the body D Mass and density of the body
32. The kinetic energy of a 100 gm bullet moving at a speed of 100 m/s is:
 A 2500J B 500 J C 1250 J D 250 J
33. Electron volt is the unit of:
 A Power B Energy C Work D Momentum
34. Which of the following is the biggest unit of energy?
 A Joule C kilo watt hour
 B watt hour D None of these
35. When the mass and speed of the body is doubled, its kinetic energy becomes:
 A 2 times B 4 times C 6 times D 8 times
36. The energy possessed by a stretched spring is:
 A Kinetic energy C Gravitational P.E
 B Electrostatic Energy D Elastic P.E
37. When a body is lifted through a height h, the work done on the body appears in the form of:
 A Kinetic energy C Electrical energy
 B Potential energy D Work
38. What is the potential energy lost by a stone of mass 1- kg when dropped from a 10 m high building?
 A Zero B 98 joules C 100 joules D 980 joules
39. Which of the following Products may not be expressed in units of energy?
 A Surface tension x area C Pressure x volume
 B Force x distance moved D Potential difference x current
40. Work done in moving a body between two points is given by
 A $W_{\text{total}} = GmM_e (1/r_1 - 1/r_n)$ C $W_{\text{total}} = GmM_e (r_n - r_1)$
 B $W_{\text{total}} = GmM_e (1/r_n - 1/r_1)$ D $W_{\text{total}} = GmM_e (r_1 - r_n)$
41. The work required to move a body from surface of earth to an infinite distance is called:
 A Absolute work done C Absolute power
 B Absolute kinetic energy D Absolute Potential energy
42. Kinetic and potential energies are interconvertible i.e. Loss kinetic energy =
 A Gain in K.E B Gain in P.E C Loss in P.E D None of these
43. The tidal energy is due to the gravitational pull of:
 A Mars on water C Sun on water
 B Moon on water D Jupiter on water

44. The energy obtained by tapping the energy of the hot molten areas inside the earth crust is:
 A Nuclear energy C Geothermal energy
 B Electrical energy D Chemical energy
45. The fossil fuels are mainly:
 A Coal and wood C Both A and B
 B Petroleum and natural gas D None of these
46. The breaking of heavy nucleus into two intermediate nucleus with the release of energy is called
 A Fission B Fusion C Both A and B D None of these
47. The energy released in the process of Fission and Fusion is called:
 A Chemical energy C Nuclear energy
 B Geothermal energy D None of these
48. Energy produce in earth's interior by radioactive decay is called:
 A Fossil energy C Geothermal energy
 B Nuclear energy D Chemical energy
49. Solar cells are thin sandwiches of:
 A Conductors B Semi-Conductors C Insulators D None of these
50. When the strings of a catapult are pulled, which energy is stored in the strings:
 A Elastic energy B Electrical energy C Nuclear energy D Chemical P.E
51. The energy stored in the chemical battery of a car is:
 A Nuclear energy C Chemical P.E
 B Electrical energy D Electrical Potential
52. When the velocity of a body is doubled:
 A Its K.E is doubled C Its momentum is doubled
 B Its P.E is doubled D Its acceleration is doubled
53. A moving body need not have:
 A Kinetic energy C Potential energy
 B Momentum D Velocity
54. If a bomb dropped from an airplane explodes in mid-air:
 A Its kinetic energy increases C Its total energy decreases
 B Its total energy increases D Its Total momentum decreases
55. The work done in lifting a mass of 50kg at a height of 2m above the ground is:
 A 0 B 25 J C 100 J D 980 J
56. A 1 kg mass has a K.E of 1 joule when its speed is:
 A 0.45 m/s B 1 m/s C 1.4 m/s D 4.4 m/s
57. A 1 kg mass has a P.E of 1 joules relative to the ground when it is at a height of:
 A 0.102 m B 1 m C 9.8 m D 32 m
58. If the force acting on a body is inversely proportional to its speed, the kinetic energy of the body is:
 A Constant
 B Directly proportional to time
 C Inversely proportional to time
 D Directly proportional to the square of time

59. Two bodies of masses m and $4m$ are dropped from the top of a tower. When they reach the ground their kinetic energy will be in the ratio:
 A 1:2 B 2:1 C 1:4 D 4:1
60. A 500g body has a velocity $v = 3\hat{i} + 4\hat{j}$ m/s at a certain instant. Its kinetic energy is:
 A 6.25 J B 62.5 J C 6.25×10^3 J D 62.5×10^3 J
61. The decrease in the potential energy of a ball of mass 20kg which falls from a height of 50 cm is:
 A 968 J B 98 J C 1980 J D None of these
62. The kinetic energy of a body of mass 2kg and momentum 2 Ns is:
 A 1J B 2 J C 3 J D 4 J



MULTIPLE CHOICE QUESTIONS

1. If a given spring constant K is cut into three identical segments, the spring constant of each segment is:
 - a. $k/2$
 - b. $k/3$
 - c. $3k$
 - d. $4k$
2. A spring 20 cm long. A force of 10N stretches through 6cm. What will be the length of spring, if a force of 20 N is applied.
 - a. 12 cm
 - b. 24 cm
 - c. 32 cm
 - d. 64 cm
3. A long spring is stretched by 2cm and its potential energy is U . If the spring is stretched by 10 cm the potential energy stored in the would be:
 - a. $U/5$
 - b. $U/25$
 - c. $25U$
 - d. $5U$
4. The period of oscillation of a mass m suspended from a spring is 2 s. If along with it another mass of 2 kg is also suspended, the period of oscillation increases by 1s. The mass m will be:
 - a. 2 kg
 - b. 1 kg
 - c. 1.6 kg
 - d. 2.6 kg
5. A particle performs simple harmonic motion of amplitude 0.02 m and frequency 2.5 Hz. What is its maximum speed? (LUMHS 2007)
 - a. 0.008 ms^{-1}
 - b. 0.050 ms^{-1}
 - c. 0.125 ms^{-1}
 - d. 0.157 ms^{-1}
 - e. 0.314 ms^{-1}
6. The length of simple pendulum having frequency 0.5 hertz is:
 - a. 1 m
 - b. 100 cm
 - c. 1000 m
 - d. All of them
7. The time period of a second pendulum is 2sec. The spherical hollow bob has a mass of a 5g. This is replaced by another solid bob of same radii but having a mass of 100g. The new time period is:
 - a. 12 sec
 - b. 6 sec
 - c. 2 sec
 - d. 8 sec
8. The acceleration of free fall on the moon is $1/6$ of that on the earth. What would be the period on the moon of a simple pendulum which has a period of 1 sec on earth (K.B 2002)
 - a. $1/6 \text{ sec}$
 - b. $1/6 \text{ sec}$
 - c. 6 sec
 - d. 6 sec
9. A transverse wave on a long horizontal rope with a wavelength of 8m travel at 2 m/s. At $t = 0$, a particular point on the rope has a vertical displacement of $+A$, where A is the amplitude of the wave. At what time will the vertical displacement of this same point on the rope be $-A$? (DUHS 2009)
 - a. $t = 1/8 \text{ s}$
 - b. $t = 1/4 \text{ s}$
 - c. $t = 1/2 \text{ s}$
 - d. $t = 2 \text{ s}$
 - e. $t = 4 \text{ s}$
10. A closed end organ pipe one metre long Resounds in its fundamental mode of frequency 100 Hz. The velocity of sound will be:
 - a. 100 m/s
 - b. 400 m/s
 - c. 330 m/s
 - d. 560 m/s
11. The ratio of fundamental frequency of an open and pipe to that of a closed end pipe:
 - a. 2:1
 - b. 1:2
 - c. 3:2
 - d. 1:1
12. The velocity if sound in air is 340m/s. The frequency of fundamental mode emitted by an open end organ pipe of length 25 cm is:
 - a. 680 hertz
 - b. 340 hertz
 - c. 250 hertz
 - d. 25 hertz

13. The frequency of the first harmonic of a string stretched between two point is 100 Hz. The frequency of third overtone is:
a. 200Hz b. 300Hz c. 400Hz d. 600Hz
14. A stretched string length 2m vibrates in 14 segments in 4 segments. The distance between consecutive node is:
a. 0.5 m b. 0.25 m c. 1.0 m d. 0.75 m
15. The fundamental frequency of a closed organ pipe is 50 Hz. The frequency of second over is:
a. 100 Hz b. 150 Hz c. 200 Hz d. 250Hz
16. The fundamental note produced by an open organ pipe has frequency n. The fundamental note produced by a closed organ pipe of the same length will have frequency.
a. $n/2$ b. $2n$ c. $n/4$ d. $4n$
17. The first overtone of an open pipe has frequency n. The first overtone of a closed pipe of the same length will have frequency.
a. $n/2$ b. $2n$ c. $3n/4$ d. $4n/3$
18. A closed organ Pipe and an open organ pipe have their overtones of the same frequency. Their length are in the ratio:
a. 1:2 b. 2:3 c. 3:4 d. 4:3
19. A vibrating string under a certain tension produces 100 vibrations per second. Find the number of vibration produced per second when the tension increased by 4 time:
a. 100 b. 300 c. 20 d. 400
20. A VHF radio station broad casts at a frequency of 90 MHz, what is the wave length of the wave broad casts by the station?
a. 0.30 m b. 3.3 m c. 12×10^{15} m d. 27×10^{15} m
21. When the temperature of an ideal gas is increased by 600K, the velocity of sound in the gas become $\sqrt{3}$ times the initial velocity at temperature $t^\circ\text{C}$. Find t.
a. -73°C b. 27°C c. 127°C d. 327°C
22. The velocity of sound in a gas is v if the pressure of gas is increased by 9 times, the speed of sound will increase by:
a. 3 times b. 9 times c. 81 times d. No change
23. The relative intensity (I/I_0) of the sound of a jet engine is 1013. The intensity level in dB will be:
a. 13 b. 130 c. 1300 d. 1.3
24. A 252 Hz, turning fork produces four beats per second when sounded with another turning fork of unknown frequency. What are the two possible values for the unknown frequency?
a. 262 Hz, 256 Hz b. 256 Hz, 248 Hz c. 249 Hz, 256 Hz d. 254 Hz, 244 Hz
25. The velocity of sound is 340 m/s, the minimum distance between the listener and the obstacle in order to hear the Echo of a sound must be:
a. 9.4 m b. 8.5 m c. 17 m d. 34 m
26. If an observer moves towards a source of sound with such a velocity that the Apparent frequency heard by him be just double the actual frequency. The velocity of an observer is (velocity of sound in air = 330 m/s):
a. 330 m/s b. 660 m/s c. 165 m/s d. 160 m/s

27. The wavelength of light coming from a star shifts towards the violet end of the spectrum. This shows that the star is:
- Stationary
 - Receding the earth
 - Approaching the earth
 - Nothing can be calculated
28. An oscillatory (vibratory) motion is always under:
- An elastic restoring force and inertia
 - An applied force
 - A periodic force
 - A gravitational force
29. When body attached with a spring has no displacement from its mean position of the body is known as:
- Extreme position
 - Equilibrium position
 - Zero position
 - Initial position
30. When a body attached with a spring has maximum displacement from its mean position. This position of the body is known as:
- Extreme position
 - Mean position
 - Equilibrium position
 - Initial position
31. Distance between mean position and extreme position is known as:
- Displacement
 - Equilibrium
 - Maximum distance
 - Amplitude
32. The force applied on a body attached with the spring is directly proportional to the:
- Amplitude
 - Displacement
 - Both of these
 - None of these
33. When a spring is stretched and then released, it moves towards the mean position due to:
- Applied force
 - Extreme force
 - Elastic restoring force
 - All of these
34. The ratio of the external force acting on a spring to increase in its length is called the:
- Applied force
 - Spring constant
 - Amplitude
 - None of these
35. If a given spring of spring constant K is cut into three identical segments, the spring constant of each segment is:
- $k/3$
 - $3k$
 - $4k$
 - k
36. In vibratory motion the:
- K.E remains constant
 - P.E remains constant
 - Total energy remains constant
 - Total momentum remains constant
37. At mean position the energy of mass attached to spring is:
- Kinetic energy
 - Potential energy
 - Wholly kinetic energy
 - Wholly potential energy
38. At mean position the potential energy of mass attached to spring is:
- Equal to PE
 - Zero
 - One
 - None of these
39. At extreme position the energy of mass attached to spring is:
- Kinetic energy
 - Potential energy
 - Wholly kinetic energy
 - Wholly potential energy

40. At extreme position the kinetic energy of mass attached to spring is:
- Equal to PE
 - Zero
 - One
 - None of these
41. The portion of wave above the mean position is called:
- Crest
 - Tough
 - Projection
 - None of these
42. The portion of wave below the mean position is called:
- Crest
 - Tough
 - Projection
 - None of these
43. What happens to the time period if the mass attached to the spring increases?
- It decreases
 - It increases
 - It remains constant
 - None of these
44. A generator produces 30 pulses in 5 seconds. Its frequency is:
- 30 Hz
 - 150 Hz
 - 6 Hz
 - 5 Hz
45. A sound wave of frequency 400Hz is traveling in air at a speed of 320m/s. The wavelength of the wave is:
- 400 m
 - 0.8 m
 - 320 m
 - 1.25 m
46. Wave transmit:
- Momentum
 - Power
 - Energy
 - Force
47. The equation for wave speed is applicable on:
- Spring waves
 - Sound waves
 - Water waves
 - All of the above
48. Which of the following does not exhibit S.H.M?
- A plucked violin string
 - A mass attached to a spring when released
 - A train shunting between two terminals
 - A simple pendulum
49. At mean position during S.H.M:
- PE is max and KE is min
 - PE is min and KE is max
 - Both PE and KE are max
 - Both PE and KE are min
50. An object undergoes SHM, its maximum speed occurs when is displacement from its mean position is:
- Maximum
 - Zero
 - Half the maximum value
 - One third of maximum value
51. Which one of the following quantities does not remain constant when a particle moves in simple harmonic motion?
- Total energy
 - Acceleration
 - Amplitude
 - Angular frequency
52. The type of vibratory motion whose instantaneous acceleration at any distance x from the mean position is proportional to x and always directed towards the mean position is called:
- Simple harmonic motion
 - Torque
 - Oscillatory motion
 - None of these

53. If a point mass m at a point P moving in a circle of radius x_0 with a constant angular velocity', where the period of oscillation is $2\pi \sqrt{m/k}$ and displacement is ' x ' then the kinetic energy of the particle is:

- a. $\frac{1}{2} kx^2$
- b. $\frac{1}{2} kx_0^2$
- c. $\frac{1}{2} k (x_0^2 - x^2)$
- d. None of these

54. The instantaneous displacement of mass attached to spring having amplitude x_0 executing SHM is:

- a. $x = x_0 \cos \sqrt{\frac{k}{m}} t$
- b. $x = x_0 \cos \sqrt{\frac{m}{k}} t$
- c. $x = x_0 \cos \sqrt{\frac{m}{k}} \times t$
- d. $x = x_0 \cos \sqrt{\frac{k}{m}} \times t$

55. The instantaneous velocity of mass attached to spring having amplitude x_0 executing SHM is:

- a. $v_0 = x_0 \sqrt{1 - \left(\frac{x^2}{x_0^2}\right)}$
- b. $v_0 = x_0 \sqrt{\frac{k}{m}} \sqrt{1 + \left(\frac{x^2}{x_0^2}\right)}$
- c. $v_0 = x_0 \sqrt{\frac{k}{m}} \sqrt{1 - \left(\frac{x^2}{x_0^2}\right)}$
- d. $v_0 = x_0 \sqrt{\frac{k}{m}} \sqrt{1 - \left(\frac{x}{x_0}\right)}$

56. The maximum velocity of mass attached to spring executing SHM is:

- a. $v_0 = x \sqrt{\frac{k}{m}}$
- b. $v_0 = x_0 \sqrt{\frac{k}{m}}$
- c. $v_0 = x_0 \sqrt{\frac{m}{k}}$
- d. $v_0 = x \sqrt{\frac{m}{k}}$

57. The total energy of mass attached to spring executing SHM is:

- a. $\frac{1}{2} kx^2$
- b. $\frac{1}{2} kx_0^2$
- c. $\frac{1}{2} kx_0$
- d. kx_0^2

58. The Time period of simple pendulum depends upon:

- a. Mass of the pendulum
- b. Material of the Pendulum
- c. Mass of the thread
- d. Length of the pendulum

59. The time period of a second pendulum is:

- a. One second
- b. Two seconds
- c. Three seconds
- d. Four seconds

60. The length of a second pendulum, where $g = 980 \text{ m/s}^2$ is:

- a. 100 cm
- b. 99.4 cm
- c. 90 cm
- d. 98 cm

61. If the length of the simple pendulum becomes four times. Its period of motion becomes:

- a. Two times.
- b. Three times.
- c. Four times.
- d. No change

62. If the mass of a simple pendulum is doubled its Time period:

- a. Become doubled
- b. Become half
- c. Become one fourth
- d. Remains constant

63. The frequency of the second pendulum is:

- a. 0.5 Hz
- b. 1 Hz
- c. 1.5 Hz
- d. 2 Hz

64. If a simple pendulum swings, which of the following quantities does not become zero throughout oscillation:

- a. Acceleration
- b. Speed
- c. Weight
- d. Momentum

65. If a simple pendulum is shifted from Karachi to Mount Everest. Its time period:

- a. Decreases
- b. Increases
- c. Remains constant
- d. Decrease slightly

66. Force responsible for vibratory motion of simple pendulum is:
 a. $mg \sin \theta$ b. $mg \cos \theta$ c. Tension d. All of them
67. For a simple pendulum the graph between L and T^2 is:
 a. Straight line b. Parabola c. Curve d. Ellipse
68. Mechanical Waves:
 a. Are longitudinal only c. Can be both longitudinal and transverse
 b. Are transverse only d. Do not require a medium for propagation
69. Of the following properties of a wave, the one that is dependent of the others is it's
 a. Velocity b. Amplitude c. Frequency d. Wave length
70. Transverse wave can propagate:
 a. Both in a gas and in a metal c. Not in a gas but in a metal
 b. In a gas but not in a Metal d. Neither in a gas nor in a metal
71. In a ripple tank, 10 waves pass across a certain point in one second. If wavelength of the waves is 4 cm, then the speed of wave is:
 a. 40 cm/sec b. 10 cm/sec c. 4 cm/sec d. 2.5 cm/sec
72. If a stone is dropped in a tank of water. The waves formed on water surface is:
 a. Longitudinal wave c. Square wave
 b. Compressional wave d. Transverse wave
73. The wave in which the particle of the medium vibrate perpendicular to the direction of wave is called:
 a. Compressional wave c. Transverse wave
 b. Longitudinal wave d. Electromagnetic wave
74. Which of the following waves are transverse wave?
 a. Sound waves c. Water waves
 b. Beaded spring waves d. Square waves
75. Ripple tank is used to study to feature of:
 a. Particle b. Wave c. Solids d. Gases
76. In a ripple tank the images of rough which appear dark on screen act as:
 a. Converging lens b. Diverging lens c. Prism d. None
77. In a ripple tank, 100 waves pass across a certain point in 1 sec. If $\lambda = 4$ the speed of wave is:
 a. 100 m/s b. 25 m/s c. 400 m/s d. 0 m/s
78. When the stationary waves are set up in a string which is fixed at both ends, the points where the amplitude is maximum are called:
 a. Antinodes c. Neither node nor antinodes
 b. Nodes d. Overtones
79. How many nodes must be there between two Antinodes?
 a. 3 b. 2 c. 1 d. Zero
80. The distance between two adjacent nodes or Antinodes in a stationary wave is:
 a. λ b. $\lambda/2$ c. $\lambda/4$ d. 2λ
81. A string at two ends vibrates in one whole segment. The standing wave pattern set up is called:
 a. First overtone c. Fundamental
 b. Second overtone d. Second Harmonics

82. Standing Waves are produced by the superposition of two identical wave moving:
- In the same direction
 - Along the same line in opposite direction
 - At right angle to each other
 - None of these
83. The movement (velocity) of particles at the anti-nodes in a stationary wave is:
- Maximum
 - Minimum
 - Zero
 - None of these
84. The wavelength of the fundamental mode of vibration of a closed end pipe is:
- $\frac{l}{2}$
 - l
 - $2l$
 - $4l$
85. Sound produced by a closed end organ pipe contains:
- All the Harmonics
 - Only odd harmonics
 - Only even harmonics
 - None of these
86. A closed end organ pipe one meter long Resounds in its fundamental mode of frequency of 100 Hz. The velocity of sound will be:
- 100m/s
 - 400m/s
 - 330m/s
 - 560m/s
87. The ratio of fundamental frequency of an open end pipe to that of a closed end pipe is:
- 2: 1
 - 1: 2
 - 3: 2
 - 1: 1
88. The point on the chord or spring which do not move at all are called:
- None
 - Antinode
 - Stationary point
 - None of these
89. The point on the chord or spring which oscillate with maximum amplitude are called:
- None
 - Antinode
 - Stationary point
 - None of these
90. The velocity of sound in air is 340m/s. The frequency of fundamental mode emitted by an open end organ pipe of length 25cm is:
- 680 Hz
 - 340 Hz
 - 250 Hz
 - 25 Hz
91. In case of stationary waves the speed of wave depends upon:
- Tension in string
 - Length and mass of string
 - Both A and B
 - None of these
92. In case of stationary waves the speed of wave is independent of:
- Length of string
 - Tension in string
 - Point from where string is plucked
 - None of these
93. If the stretched string is plucked from one quarter of its length from one end stationary waves will be there vibrating it in:
- One loop
 - Two loops
 - Three loops
 - Four loops
94. Harmonics produced in a stretched string are:
- Even in number
 - Odd in number
 - Sometimes odd sometimes even
 - None of these
95. Turning fork is an instrument to produce:
- Heat
 - Energy
 - Light
 - Sound
96. The frequency of the first harmonic of a string stretched between two points is 100 Hz the frequency of the third overtone is:
- 200 Hz
 - 300 Hz
 - 400 Hz
 - 600 Hz

97. A stretched string of length 2 m vibrates in 4 segments. The distance between consecutive nodes is:
- 0.5 m
 - 0.25 m
 - 1.0 m
 - 0.75 m
98. The fundamental frequency of a closed organ pipe is 50 Hz. The frequency of the second overtone is:
- 100 Hz
 - 150 Hz
 - 200 Hz
 - 250 Hz
99. The fundamental not produced by an open organ pipe has frequency n . The fundamental not produced by a closed organ pipe of the same length will have frequency:
- $n/2$
 - $2n$
 - $n/4$
 - $4n$
100. The first overtone of an open pipe has frequency n . The first overtone of a closed pipe of the same length will have frequency:
- $n/2$
 - $2n$
 - $3n/4$
 - $4n/3$
101. A closed organ pipe and an open organ pipe have their first overtones of the same frequency. Their lengths are in the ratio:
- 1:2
 - 2:3
 - 3:4
 - 4:3
102. With the increase in temperature, the frequency of the sound from an organ pipe:
- Decreases
 - Increases
 - Remains unchanged
 - Change erratically
103. Speed of sound in air is 300 m/s, the distance between two successive nodes of a stationary wave of frequency 1000 Hz is:
- 10 cm
 - 15 cm
 - 20 cm
 - 30 cm
104. If the speed of sound is v , then the frequency of the fundamental note emitted by a closed pipe of length l is:
- $v/2l$
 - $v/4l$
 - $3v/4l$
 - $4v/l$
105. The frequency of the fundamental note emitted by string of length l , clamped at both ends is (v is the velocity of waves in string):
- v/l
 - $2v/l$
 - $v/2l$
 - $v/4l$
106. Standing Waves can be produce:
- On a string clamped at both the ends.
 - On a string clamped at one end and free at the other
 - when incident waves get reflected from a wall
 - All of these
107. The velocity of sound in air is 330 m/s. The fundamental frequency of an organ pipe, open at both ends and of length 0.3 m will be:
- 200 Hz
 - 550 Hz
 - 300 Hz
 - 275 Hz
108. Stationary waves of frequency 300 Hz are formed in a medium in which the velocity of sound is 1200 m/s. The distance between a node and the neighboring anti-node is:
- 1 m
 - 2 m
 - 3 m
 - 4 m
109. In a stationary longitudinal wave, nodes are points of:
- Maximum pressure
 - Minimum pressure
 - Minimum pressure variation
 - Maximum pressure variation

110. "Stationary waves" are so called because in them:
- The particles of the medium are not disturbed at all
 - The particles of the medium do not execute SHM
 - There occurs no flow of energy along the wave
 - The interference effect can't be observed
111. Which of the following is a practical application of stretched strings for vibrating a portion of the string into desired number of loops?
- Sonometer
 - Lactometer
 - Hydrometer
 - Thermometer
112. All the laws of transverse vibration of string can be verified using sonometer, which of the following Law is incorrect:
- $f \propto 1/L$
 - $f \propto \sqrt{T}$
 - $f \propto 1/\sqrt{T}$
 - $f \propto 1/T$
113. With the propagation of a longitudinal wave through a material medium, the quantities transferred in the direction of propagation are:
- Energy, momentum and mass
 - Energy and momentum
 - Energy and mass
 - Mass
114. The types of waves produced in a sonometer wire:
- Transverse progressive
 - Transverse stationary
 - Longitudinal progressive
 - Longitudinal stationary
115. A sonometer wire is vibrating in the second overtone. In the wire there are:
- Two nodes and two Antinodes
 - One node and two Antinodes
 - Four nodes and three Antinodes
 - Three nodes and three Antinodes
116. Medium air does not possess modulus of rigidity, so waves produced in air are:
- Transverse
 - Longitudinal
 - Stationary
 - None of these
117. A string of length l . Fixed at the two ends. Is plucked from the middle position so as to vibrate in one loop. The wavelength of the wave in this mode is equal to:
- $l/2$
 - l
 - $3l/2$
 - $2l$
118. Under a given tension T , a string of length l vibrates in one second. To make it vibrate in two segments with the same frequency, the tension should be:
- $T/2$
 - $T/4$
 - $2T$
 - $4T$
119. The velocity of longitudinal waves in the metallic rod depends upon its:
- Bulk modulus
 - Rigidity
 - Young's modulus
 - Stress only
120. The waves in which the various particles of the medium vibrate about their mean position along the direction of waves are called:
- Longitudinal wave
 - Stationary wave
 - Transverse wave
 - Surface wave
121. The increasing velocity of sound in air in 1°C rise in temperature is:
- 0.61 m/s
 - 61 m/s
 - 1.61 m/s
 - 2 m/s
122. Velocity of sound in a medium depends upon:
- Density of medium
 - Modulus of elasticity of medium
 - Both density and modulus of elasticity
 - Pressure

123. Speed of sound in air is independent of:
 a. Temperature b. Viscosity c. Density d. All of them
124. The speed of sound in air at 0°C is:
 a. Zero b. 780 m/s c. 332 ft/sec d. 332 m/sec
125. When soundwave enters from air into water, which of the following quantity remains unchanged?
 a. Wavelength b. Velocity c. Frequency d. Amplitude
126. Speed of sound in air (fluid):
 a. Increase with temperature c. Remains unchanged
 b. Decrease with temperature d. None of these
127. Sound Travels faster:
 a. In cold medium than in hot medium c. Remains unchanged
 b. In hot medium than in cold medium d. None of these
128. If the air pressure is doubled, the speed of sound:
 a. Decreases by 0.61 m/s c. Increases by 61 cm/s
 b. Increases by 1.41 times d. Remains unchanged
129. The quantities which together determine the speed of sound in a liquid are:
 a. The bulk modulus and the density c. The young modulus and density
 b. The bulk modulus and the pressure d. The temperature and the pressure
130. The value of speed of sound was first determined by:
 a. Huygen b. Laplace c. Newton d. None of these
131. The relation for the velocity of sound in gas is:
 a. $v = \sqrt{P/\rho}$ b. $v = \sqrt{\rho P/\rho}$ c. $v = \rho P/\gamma$ d. $v = \sqrt{\gamma P/\rho}$
132. The Speed of sound is:
 a. Always constant c. Varies with height
 b. Varies with time d. Varies with medium
133. No sound can be heard in:
 a. Water b. Metals c. Glass d. Vacuum
134. Sound Travels fastest in:
 a. Gas b. Air c. Water d. Vacuum
135. Choose the correct statement:
 a. Sound waves are transverse waves
 b. Sound travels fastest through vacuum
 c. Sound travels faster in solids than in gases
 d. Sound travels faster in gases than in Liquids
136. The temperature at which the speed of sound in air becomes double its value at 0°C is:
 a. 1092°C b. 819 K c. 819°C d. 546°C
137. Laplace's Correction in the formula for the speed of sound given by Newton was needed because sound wave:
 a. Are longitudinal c. Propagate Adiabatically
 b. Propagate Isothermally d. Have long wavelengths

138. Speed of sound in gas is proportional to:
- Square root of Isothermal elasticity
 - Square root of Adiabatically elasticity
 - Isothermally elasticity
 - Adiabatic elasticity
139. The speed of sound in air at STP is 300 m/sec. If the air pressure becomes double, the temperature remaining the same, the speed of sound would become:
- 1200 m/s
 - 600 m/s
 - $300\sqrt{2}$ m/s
 - 300 m/s
140. The velocity of sound is generally greater in solids than in gases because:
- The density of solids is high and the elasticity is low
 - Both the density and the elasticity of solids are very low
 - The density of solids is low and the elasticity is high
 - The elasticity of solids is very high
141. Frequencies less than 20 Hz are called:
- Ultrasonic
 - Supersonics
 - Infrasonic
 - None of these
142. Frequency is more than 20000 Hz are called:
- Ultrasonic
 - Supersonics
 - Infrasonic
 - None of these
143. Ultrasonic have The frequencies:
- In the audible range
 - Above the upper limit of audible range
 - Below the lower limit of audible range
 - None of these
144. An average human ear can hear the sound of frequency:
- 200 to 500 Hz
 - 20 to 20,000 Hz
 - 30,000 to 50,000
 - Above 50,000 Hz
145. Decibel (or bel) is the unit of:
- Energy
 - Intensity
 - Intensity level
 - Power
146. The intensity level of ordinary conversation is:
- Zero
 - 60 db
 - 120 db
 - Infinite
147. The intensity level of faintest audible sound is:
- Zero
 - 60 db
 - 100 db
 - 120 db
148. Pitch of sound determines:
- Quality of sound
 - Softness or graveness
 - Loudness of sound
 - Intensity level
149. Notes played on piano and violin are distinguished because of their:
- Pitch
 - Loudness
 - Quality
 - Quality and pitch
150. The energy transmitted per second through a unit area by the sound wave is called:
- Propagation of sound waves
 - Energy of sound waves
 - Intensity of sound waves
 - Reflection of sound waves
151. The loudness L is proportional to the logarithm of intensity I is called:
- Newton's formula
 - Weber-Fechner Law
 - Laplace's formula
 - None of these
152. Intensity level in decibels of intensities I and I_0 is given by:
- $10 \log_{10} (I/I_0)$
 - $10 \ln (I/I_0)$
 - $10 \ln (I_0/I)$
 - $10 \log_{10} (I_0/I)$

153. Unit for intensity of sound is:
 a. J/s b. W/s^2 c. W/m^2 d. N/m^2
154. One bel =
 a. 1000 dB b. 100 dB c. 10 dB d. 1 dB
155. The loudness of sound depends upon:
 a. Frequency of sound c. Wavelengths of sound
 b. Intensity of sound d. Distance from source
156. We can distinguish between two sounds of same loudness and pitch due to:
 a. Quality of sound c. Wavelength of sound
 b. Intensity of sound d. Intensity of ear
157. The pitch of sound depends upon:
 a. Loudness c. Wavelength
 b. Frequencies d. Distance from source
158. The sounds which produces a pleasing sensation on the air are called:
 a. Music b. Noise c. Pleasing sounds d. None of these
159. The sound which produce a jarring or displeasing effect are called:
 a. Music b. Noise c. Pleasing sounds d. None of these
160. Ultrasonic waves are those waves which:
 a. Human being cannot hear c. Have high velocity
 b. Human being can hear d. Have large amplitude
161. The quality of sound produced by an instrument depends on the
 a. Frequency c. Number of Overtones
 b. Intensity d. None of these
162. The same notes been played on Sitar and Veena differ in:
 a. Quality c. Both quality and pitch
 b. Pitch d. Neither quality nor pitch
163. Decibel is a:
 a. Musical note c. Unit of intensity of intensity
 b. Musical instrument d. Unit of intensity level
164. When we hear a sound we can identify its source from:
 a. The wavelength of sound c. The intensity of sound
 b. The overtones present in the sound d. The amplitude of sound
165. When two or more waves in the same medium travel the next displacement of the medium caused by the resultant wave at any point is equal to the algebraic sum of the displacements of all the wave. The above principle is called:
 a. Superposition b. Acoustics c. Beats d. Doppler's Effect
166. Beats occur because of:
 a. Interference b. Reflection c. Refraction d. Doppler effect
167. When we are produced by two waves of nearly the same frequency?
 a. The Beat frequency depends on the position where the Beats are Heard
 b. The Beat frequency decreases as time passed

- c. the particles vibrate simple harmonically with a frequency equal to difference of the two frequencies
- d. the amplitude of vibration at any point changes simple harmonically with a frequency equal to the difference of the two frequencies
168. If two waves of the same frequency but different amplitudes superpose at a point:
- There will be no interference
 - The intensity of sound at any point will vary with time
 - There will be interference but the minimum intensity will not be zero
 - There will be interference with minimum intensity is equal to zero
169. When two sound waves are superimpose, beats are produced when they have
- Different amplitudes and phases
 - Different velocities
 - Different phase
 - Different frequencies
170. The periodic increase and decrease in loudness of sound is called:
- Resonance
 - Silence zone
 - Diffraction
 - Beats
171. When two tuning Forks of slightly different frequencies are sounded simultaneously, then the number of waves produced per second is equal to:
- Sum of their frequencies
 - Product of their frequencies
 - Difference of their frequencies
 - Ratio of their frequencies
172. Two tuning forks of frequencies 265 Hz and 262 Hz are sounded together. The number of beats produced per second is:
- Zero
 - 3
 - 527
 - 1.01
173. Overtones are:
- Nothing but beats
 - The fundamental produced in an organ pipe
 - The notes produced other than fundamental
 - None of these
174. The beats are used to find out:
- Unknown frequencies
 - Turning musical instruments
 - Both A and B
 - None of these
175. The change in pitch of sound caused by the relative motion of either the source of sound or the listener is called:
- Acoustics
 - Doppler's Effect
 - Laplace's Correction
 - None of these
176. The study of production and properties of sound is called:
- Doppler's Effect
 - Acoustics
 - Reverberation
 - Silence zone
177. The persistence of sound in an auditorium is called:
- Resonance
 - Acoustics
 - Reverberation
 - None of these
178. When a source of sound waves towards stationary listener. It causes:
- A decrease in frequency
 - An increase in frequency
 - An increase in wavelength
 - No change in frequency

179. Doppler's Effect can be applied to:
- Light waves only
 - Sound waves only
 - Both light and sound waves
 - Neither light nor sound waves
180. Silence zone is due to:
- Constructive interference
 - Destructive interference
 - Resonance
 - Doppler's Effect
181. Factors affecting acoustics of halls adversely are:
- Echo
 - Reverberation
 - Focusing of sound at certain spot
 - All of these
182. The listener moving towards a source at rest:
- Detect a large frequency
 - Detects a low pitch
 - Detects a higher pitch
 - Both a & c
183. The source moves away from the stationary listener, the wavelength of the sound arriving at the listener is:
- Greater
 - Smaller
 - Zero
 - Infinity
184. Which of the following are the applications of Doppler's Effect?
- Tracking of satellites
 - Measuring the speed of automobile
 - Radio detection and ranging
 - All of these



MULTIPLE CHOICE QUESTIONS

1. In a double slit experiment, eight fringes occupy 2.4 mm on a screen 150 mm away from the slits. The wavelength of light is 500 nm. Find the slit separation.
 a. $2.5 \mu\text{m}$ b. $250 \mu\text{m}$ c. $25 \mu\text{m}$ d. None
2. When the light from two lamps falls on a screen, no interference pattern can be obtained. Why is this? (LUMHS 2007)
 a. The lamps are not point sources
 b. The lamps emit light of different amplitudes
 c. The light from the lamps is not coherent
 d. The light from the lamps is white
3. What happens, if a monochromatic light is incident on a very narrow slit?
 a. Polarization b. Interference c. Diffraction d. None of these
4. How much moveable mirrors should be moved to the place a bright fridge by a dark fringe
 a. $\lambda/2$ b. $\lambda/4$ c. 2λ d. None of these
5. How much should the movable mirror of the Michelson's interferometer be moved in order to observe 400 fringes with reference to a point? The wavelength of the light used is 600°A .
 a. $12 \mu\text{m}$ b. $1.2 \mu\text{m}$ c. $120 \mu\text{m}$ d. None of these
6. In Newton's rings when interference pattern is viewed from above by means of reflected light, the central spot appears
 a. Bright b. Dark c. Blue d. Red
7. In a spectrometer experiment, monochromatic light is incident normally on a diffraction grating having 4.5×10^5 lines per meter. The second-order lines is seen at an angle of 30° to the normal. What is the wavelength of the light (approx.)? (LUMHS 2007)
 a. 200 nm b. 430 nm c. 500 nm d. 556 nm
8. In 1676, who showed that a wave theory of light explains the laws of reflection, refraction and the phenomena of double refraction.
 a. Christian Huygens c. Isaac Newton
 b. Thomas young d. James Clark Maxwell
9. First real demonstration that light is a wave phenomenon was made:
 a. Christian Huygens c. Isaac Newton
 b. Thomas young d. James Clark Maxwell
10. In 1873, who showed that electromagnetic waves can propagate through vacuum?
 a. Christian Huygens c. Isaac Newton
 b. Thomas young d. James Clark Maxwell
11. Which of the following is not electromagnetic in nature?
 a. Infrared rays b. Ultraviolet rays c. Radar waves d. Sound waves
12. In 1905, who explained the photoelectric effect.
 a. Albert Einstein c. A.H Compton
 b. Isaac Newton d. James Clark Maxwell

13. The photoelectric effect and Compton effect supported the:
 - a. Quantum theory of light
 - b. Wave theory of light
 - c. Both A and B
 - d. None of these
14. Interference diffraction and polarization can be explained on the basis of:
 - a. Corpuscular theory of light
 - b. Wave theory of light
 - c. Both A and B
 - d. None of these
15. The velocity of light in vacuum or air is:
 - a. 3×10^{10} m/s
 - b. 3×10^8 m/s
 - c. 0.3×10^8 m/s
 - d. 3×10^{-8} m/s
16. Photoelectric effect is explaining if light is considered to travel in the form of:
 - a. Waves
 - b. Particles
 - c. Both waves and particles
 - d. None of these
17. Which of the following phenomena cannot be explained on Wave nature of light?
 - a. Interference
 - b. Diffraction
 - c. Polarization
 - d. Compton effect
18. Which of the following phenomena cannot be explained on particle nature of light?
 - a. Photoelectric effect
 - b. Interference
 - c. Compton effect
 - d. Pair Production
19. Modern view about nature of light is that light behaves as:
 - a. Waves only
 - b. Particles only
 - c. Both waves and particles
 - d. None of these
20. The medium in which speed of light is the same in all directions is called:
 - a. Homogeneous
 - b. Non-homogeneous
 - c. Empty
 - d. None of these
21. Wavelength of X rays is of the order of:
 - a. $1 \text{ \AA}$
 - b. $10 \text{ \AA}$
 - c. $100 \text{ \AA}$
 - d. $1000 \text{ \AA}$
22. Which of the following has greater frequency?
 - a. Ultraviolet rays
 - b. Visible rays
 - c. Infrared rays
 - d. Gamma rays
23. The focus of all points having the same phase of vibration is known as:
 - a. Amplitude
 - b. Trough
 - c. Crest
 - d. Wave front
24. A point source of light placed in a homogeneous medium produces a:
 - a. Plane wave front
 - b. Spherical wave front
 - c. Cylindrical wave front
 - d. Rectangular wave front
25. The wave front formed at a very large distance from a line sources is:
 - a. Cylindrical wave front
 - b. Spherical wave front
 - c. Plane wave front
 - d. Elliptical wave front
26. Huygens's principle is applied to:
 - a. Locate the wave front
 - b. Explain Diffraction
 - c. Explain interference
 - d. Find out the speed of light
27. Huygens's principle tells us that:
 - a. Light has dual nature
 - b. Light travels in straight line
 - c. All point on primary wave front are source of secondary wave front
 - d. None of these

28. A phenomenon in which two waves sport each other at some points and cancel each other at some other points is known as:
- Interference
 - Diffraction
 - Polarization
 - Dispersion
29. Which of the following phenomena is the result of superposition of two wave?
- Interference
 - Diffraction
 - Polarization
 - Dispersion
30. To obtain interference of light waves:
- The sources should be phase coherent
 - The sources should be monochromatic
 - The principle of superposition should be applicable
 - All of the above
31. The phenomenon of interference is shown by:
- Longitudinal mechanical waves only
 - Transverse mechanical waves only
 - Non-mechanical transverse waves only
 - All of the above types of waves
32. Waves nature of light follows because:
- Light rays travel in a straight line
 - Light exhibit the phenomenon of reflection and refraction
 - Light exhibit the phenomenon of interference
 - Light causes the phenomenon of Photoelectric effect
33. Young's double slit experiment explains:
- Diffraction of light
 - Interference of light
 - Polarization of light
 - None of these
34. In young's double slit experiment, the condition for constructive interference is:
- $2d \sin \theta = m\lambda$
 - $d \sin \theta = m\lambda$
 - $d \sin \theta = (m + \lambda/2)$
 - $d \sin \theta = (m + 1/2) \lambda$
35. For Destructive interference the path difference between two light rays is equal to:
- Zero
 - $m\lambda$
 - $(m + 1/2) \lambda$
 - None of these
36. In Young's double slit experiment, the width (or fringe spacing) is equal to:
- $\lambda d/L$
 - $d/\lambda L$
 - $2\lambda d/L$
 - $\lambda L/d$
37. The positions of bright fringes in young's double slit experiment a given by:
- $Yd = (\lambda L/d) m$
 - $Yd = \lambda d/LM$
 - $Yd = mLd/\lambda$
 - $Yd = mLd\lambda$
38. The position of dark fringes in young's double slit experiment are given by:
- $Yd = \lambda L/d (m + 1)$
 - $Yd = (\lambda L/d) m$
 - $Yd = \lambda d/L (m + 1/2)$
 - $Yd = \lambda L/d (m + 1/2)$
39. Distance between two consecutive bright fringes (or dark fringes) in young's double slit experiment is called:
- Wavelength
 - Fringe spacing
 - Half-wavelength
 - Width of slit
40. In an interference pattern:
- Bright fringes are wider than dark fringes
 - Dark images are wider than bright fringes
 - Both dark and bright fringes are of equal width
 - None of these

41. In young's experiment for interference of light with two slits, maxima occur at angle for $\sin \theta = m\lambda/d$ here d is:
- Distance of slits from screen
 - Distance between dark and bright fringes
 - Distance between slits
 - Width of m^{th} fringe
42. Young's experiment established that:
- Light consists of waves
 - Light consists of particles
 - Light is neither particle nor waves
 - Light is both particle and wave
43. Common examples of thin film are:
- Soap bubbles
 - Thin layer of oil floating on water
 - A and B both
 - None of these
44. The appearance of colors when white light falls on thin films is due to:
- Interference
 - Dispersion
 - Diffraction
 - Refraction
45. When a Ray of light passes from rarer to denser medium it suffers a phase change of:
- 90°
 - 180°
 - 60°
 - 70°
46. Condition for constructive interference in thin films is:
- $2nt = m\lambda$
 - $2nt = (m + 1/2)\lambda$
 - $2nt = (m/2)\lambda$
 - $2nt = (m/3)\lambda$
47. Condition for destructive interference in thin films is:
- $2nt = m\lambda$
 - $2nt = (m + 1/2)\lambda$
 - $2nt = (m/2)\lambda$
 - $2nt = (m/3)\lambda$
48. Central spot in Newton's rings pattern is:
- Dark
 - Bright
 - Sometimes dark and sometimes light
 - All of these
49. Michelson interferometer is an instrument used to measure:
- Frequency of light
 - Intensity of light
 - Amplitude of light
 - Wavelength of light
50. The fringes in Michelson interferometer are formed due to:
- Diffraction
 - Interference
 - Polarization
 - None of these
51. Though quantum theory of light explain the number of phenomena observed with light, it is necessary to retain the wave nature of light to explain the phenomenon of:
- Photoelectric effect
 - Diffraction
 - Compton effect
 - Black body radiation
52. To observe diffraction, the size of the obstacle:
- Should be of the same order as the wavelength
 - Should be much larger than the wavelength
 - Has no relation to wavelength
 - Should be exactly half the wavelength
53. Bright colors exhibited by spider's web, exposed to sunlight are due to:
- Interference
 - Resolution
 - Diffraction
 - Polarization

54. A beam of monochromatic light of wavelength λ falls normally on a diffraction grating of grating element d . If θ is the angle between the second order diffracted beam and the direction of the incident light. What is the value of $\sin \theta$?
- a. λ/d b. d/λ c. $2\lambda/d$ d. $2d/\lambda$
55. Light of wavelength λ is incident normally on a diffraction grating for which the slit spacing is equal to 3λ . What is the sine of the angle between the second order maximum and the normally?
- a. $1/6$ b. $1/3$ c. $2/3$ d. $3/2$
56. According to Bragg's law:
- a. $d \sin \theta = (m/3) \lambda$ c. $d \sin \theta = m \lambda$
b. $2d \sin \theta = m \lambda$ d. $d \sin \theta = (m + 1/2) \lambda$
57. Which of the following demonstrate that light waves are transverse is called?
- a. Interference b. Polarization c. Diffraction d. Reflection
58. Which one of the following cannot be polarized?
- a. X-rays b. Light rays c. Water waves d. Sound wave
59. Longitudinal waves do not exhibit (show):
- a. Diffraction b. Polarization c. Interference d. Refraction



MULTIPLE CHOICE QUESTIONS

1. If an object is placed in front of two plane mirrors arranged at right angles to each other, the number of images seen will be:
 a. 4 b. 3 c. 2 d. 1
2. An object is placed 60cm in front of a concave spherical Mirror whose focal length is 40cm. Which of the following best describes the image? (DUHS 2009)

Nature of Image	Distance From Mirror
a. Virtual	24 cm
b. Real	24 cm
c. Virtual	120 cm
d. Real	120 cm
e. Real	240 cm
3. An object is placed 60 cm from a spherical convex mirror. If the mirror forms a virtual image 20 cm from the mirror, what is the magnitude of the mirror's radius of curvature? (DUHS 2009)
 a. 7.5 cm c. 30 cm
 b. 15 cm d. 60 cm
 e. 120 cm
4. The speed of light in a transparent material is 0.75 times its speed in air. The absolute index of refraction of the material is approximately.
 a. 1.3 b. 0.75 c. 2.3 d. 4.0
5. The size of the image compared to the size of the object will be approximately:
 a. Twice as large c. 1.5 times as large
 b. More than twice as large d. Smaller
 e. The same size
6. If the power of a converging lens is 4 diopters what is the focal length of lens:
 a. 20 cm b. 25 cm c. 10 cm d. 50 cm
7. Two lenses have focal length 20 cm and -40cm. The focal length of combination in contact is: (LUMHS 2007)
 a. +20 cm b. +40 cm c. -20 cm d. -40 cm
8. By using adjustable aperture of a lens we can reduce the defect of the lens which is called:
 a. Astigmatism c. Spherical aberration
 b. Chromatic aberration d. None of them
9. The magnifying power of a lens of focal length 25 cm is:
 a. $\frac{1}{2}$ b. 1 c. Zero d. 2
10. If magnifying power of simple microscope is 6 the focal length of the lens used is:
 a. 6 cm b. 5 cm c. 25 cm d. - 5 cm
11. The magnifying power of a telescope (Astronomical) is 8 and the distance between two lenses is 54 cm. The focal length of objective and eyepiece are respectively. _____ cm. (LUMHS 2007)
 a. 6, 48 b. 48, 6 c. 8, 64 d. 64, 8

12. In the terrestrial telescope the central lens is used to:
- Erect the image
 - Increase magnifying power
 - Both of these
 - None of these
13. A lens which converges a beam of parallel rays to a point is called:
- Concave lens
 - Convex lens
 - Plano concave lens
 - Plano convex lens
14. A lens which diverges a beam of parallel rays is called:
- Plano convex lens
 - Convex lens
 - Concave lens
 - None of these
15. The converging lens is:
- Thicker in centre and thinner at ends
 - Thicker at ends and thinner at centre
 - Thick at both centre
 - Thin at both centre
16. The diverging lens is:
- Thicker in centre and thinner at ends
 - Thicker at ends and thinner at centre
 - Thick at both centre
 - Thin at both centre
17. A point at which the incident parallel rays of light converge or appear to diverge after passing through a lens is called:
- Centre of curvature
 - Pole
 - Aperture
 - Principal focus
18. The distance between focus point and the optical Centre is called:
- Principal axis
 - Aperture
 - Focal length
 - Radius of curvature
19. A straight line joining the centre of curvature of two surfaces of lens is:
- Optical centre
 - Principal axis
 - Aperture
 - Principal focus
20. The diameter of lens is called:
- Principal axis
 - Optical centre
 - Aperture
 - Principal focus
21. The fixed point lying inside or outside on its Axis such that a Ray of light passing through it suffers no deviation is called:
- Principal focus
 - Principal axis
 - Optical centre
 - None of these
22. The focal length of a convex lens is:
- Negative
 - Positive
 - Either positive or negative
 - None of these
23. The focal length of a concave lens is:
- Negative
 - Positive
 - Neither negative nor positive
 - None of these
24. The reciprocal of focal length of a lens is expressed in meters is called:
- Focal length
 - Aperture
 - Power of a lens
 - Principal axis
25. The unit of power of a lens is:
- Newton
 - Watt
 - Horse power
 - Diopter
26. The power of a lens of focal length 100 cm is:
- 1 diopter
 - 2 diopter
 - 10 diopter
 - 100 diopter

27. If an object is placed between the focus and a converging lens, the final image will be:
- Real inverted and magnified in front of the lens
 - Real erect and magnified behind the lens
 - Virtual erect and magnified in front of the lens
 - None of these
28. A convex lens forms a virtual image only when the object lies:
- Between f and $2f$
 - Beyond $2f$
 - At the principal focus
 - Between focus and optical centre
29. If an object is placed between f and $2f$ of a convex lens, the image will be:
- Real inverted and magnified
 - Real erect and magnified
 - Virtual erect and diminished
 - Virtual erect and magnified
30. A double convex lens act as a diverging lens when the object is:
- Between f and $2f$
 - At the focus
 - Within the focus
 - Beyond $2f$
31. Linear magnification is defined as the ratio of:
- Size of object to size of image
 - Size of image to size of object
 - Size of image to focal length
 - None of these
32. If the object is beyond $2f$ in converging lens then image is:
- Real inverted and diminished
 - Real erect and small size
 - Virtual erect and magnified
 - Virtual erect and small size
33. If the object is at $2f$ in converging lens then image is:
- Real inverted and diminished
 - Real inverted and same size
 - Virtual erect and magnified
 - Virtual inverted and small size
34. If the object is inside the focus in converging lens then image is:
- Real inverted and diminished
 - Real erect and small size
 - Virtual erect and magnified
 - Virtual inverted and small size
35. The power of a lens is 4 diopters. Its focal length is:
- 20 cm
 - 25 cm
 - 50 cm
 - 400 cm
36. The focal length of a convex lens is 50 cm. Its power is:
- +5 D
 - 50 D
 - +2 D
 - 2 D
37. Two lenses of powers + 12 and - 2D are in contact. The focal length of the combination is:
- 10 cm
 - 12.5 cm
 - 16.6 cm
 - 8.33 cm
38. Are convex lens of focal length 40 cm is in contact with a concave lens of focal length 25cm. The power of the combination in diopters is:
- 1.5
 - 6.5
 - +6.5
 - +6.67
39. Economics lens of focal length 10 cm is placed in contact with a concave lens is focal length 20cm. The nature and focal length of the combination is:
- Diverging, 20 cm
 - Diverging, 10cm
 - Converging, 20 cm
 - Converging, 10 cm
40. Defects found in lenses are:
- Chromatic aberration only
 - Spherical aberration only
 - Chromatic and Spherical aberration only
 - Distortion

41. The inability of the lens to focus all the dispersed colors at one point is called:
- Distortion
 - Spherical aberration
 - Chromatic aberration
 - None of these
42. The defects of chromatic aberration can be removed by using:
- An additional convex lens
 - A combination of a convex and a concave lens
 - Two concave lens
 - Two convex lens
43. Lenses of large aperture suffer from the defect of:
- Spherical aberration
 - Chromatic aberration
 - Neither spherical nor chromatic aberration
 - None of these
44. A long sighted person cannot see clearly:
- Distant objects
 - Nearer object
 - Both distant and nearer objects
 - None of these
45. In the case of hypermetropia:
- The image of a near object is formed behind the retina
 - A Convex Lens be used for correction
 - The image of a distant object is formed in front of the retina
 - Both a and b
46. The far point of a myopic eye is 250 cm. The connecting lens should be a:
- Diverging lens of focal length 250 cm
 - Converging lens of focal length 250 cm
 - Diverging lens of focal length 125 cm
 - converging lens of focal length 125 cm
47. Spherical aberration in a thin lens can be reduced by:
- Using a monochromatic light
 - Using a doublet combination
 - Using a circular annular mask over the lens
 - Increasing the size of the lens
48. Chromatic aberration in a lens is caused by:
- Reflection
 - Interferences
 - Diffraction
 - Dispersion
49. An achromatic combination is to be made using a convex and a concave lens. The two lenses should have:
- Their Power equal
 - Their refractive indices equal
 - Their dispersive Powers equal
 - The product of their power and dispersive powers equal
50. All of the following statements are correct except:
- The magnification produced by a convex mirror is always less than 1
 - A virtual erect same size image can be obtained by using a mirror
 - A virtual, erect magnified image can be formed by using a concave mirror
 - A real inverted same size image can be formed by using a convex mirror
51. Magnifying power of a simple microscope is:
- $M = 1 - d/f$
 - $M = 1 + f/d$
 - $M = 1 + d/f$
 - $M = d/f - 1$

52. When image is virtual and erect the magnification is:
 a. Positive b. Negative c. Maximum d. Minimum
53. The compound microscope consists of:
 a. Two concave lens c. A concave and a convex lens
 b. Two convex lenses d. None of these
54. In compound microscope the focal length of:
 a. Objective is greater than eyepiece c. Both are of equal focal length
 b. Eyepiece is greater than objective d. Lenses of any focal lengths can be used
55. The final image seen by the eye through the microscope is
 a. Real and magnified c. Virtual and magnified
 b. Real and diminished d. Virtual and diminished
56. The magnifying power of a compound microscope is:
 a. $\frac{L}{f_o(1+\frac{d}{f_e})}$ c. $\frac{f_o}{L(1+\frac{d}{f_e})}$
 b. $\frac{L}{f_o(1+\frac{f_e}{d})}$ d. $\frac{f_o}{L(1+\frac{f_e}{d})}$
57. For high magnification in a compound microscope, which of the following should have short focal length?
 a. Objective lens c. Both objective and eyepiece
 b. Eyepiece lens d. None of these
58. The ratio of angle subtended at eye of an object when viewed through lens to the angle formed by object on an unaided eye is called:
 a. Linear magnification c. All of these
 b. Magnifying power d. None of these
59. In Compound microscope, the intermediate image is:
 a. Visual erect and magnified c. Real inverted and magnified
 b. Real erect and magnified d. Virtual erect and reduced
60. The minimum number of lenses used in a telescope is:
 a. One b. Two c. Three d. Four
61. For high magnification of astronomical telescope the objective lens should be of:
 a. Large focal length c. Equal focal length
 b. Short focal length d. None of these
62. Telescope is the instrument used to obtain as clear and distinct image of an object.
 a. Near to the observer c. Very far of the observer
 b. Of Very small size d. All of the above
63. The objective lens used in a telescope is of:
 a. Large focal length and diameter c. Large focal length and short diameter
 b. Short focal length and diameter d. Short focal length and long diameter
64. The length of astronomical telescope in normal adjustment is:
 a. $f_o + f_e$ b. $f_o - f_e$ c. $f_o \times f_e$ d. f_o / f_e

65. The magnifying power of a telescope is defined as the ratio of the angle subtended by the image at the eye as seen through the telescope to the angle subtended by the image at the:
- Objective
 - Eye
 - Eyepiece
 - None of these
66. The angle magnification of a telescope is given mathematically as:
- $M = 1 + f_o/f_e$
 - $M = f_o/f_e + 1$
 - $M = f_e/f_o$
 - $M = f_o/f_e$
67. Which one of the following has total length when set for parallel rays?
- Terrestrial Telescope
 - Astronomical telescope
 - Galilean telescope
 - None of these
68. Astigmatism for human eye can be removed by using:
- Concave lens
 - Convex lens
 - Cylindrical lens
 - Prismatic lens
69. The human eye has a lens which has a:
- Soft portion at its centre
 - Hard surface
 - Varying refractive index
 - Constant refractive index
70. Two connect lenses of focal length 0.3 m and 0.05 m are used to make a astronomical telescope the distance kept between them is equal to:
- 0.35 m
 - 0.25 m
 - 0.175 m
 - 0.15 m
71. Length of a Galilean telescope in normal adjustment in terms of the focal length of the objective f_o and that of the eyepiece f_e is:
- f_o/f_e
 - $f_e - f_o$
 - $f_o - f_e$
 - $f_o + f_e$
72. A Convex Lens of focal length 16 cm forms a real image double the size of the object. The distance of the object from the lens is:
- 32 cm
 - 24 cm
 - 20 cm
 - 8 cm
73. A combination of two thin lenses in contact behaves as a converging lens of focal length 80 cm. If one of the lenses is a converging lens of focal length 20cm, the power of the other lens is:
- 1.66 D
 - 4.00 D
 - 1.00 D
 - 3.75 D
74. The focal length of the objective and eyepiece of an astronomical telescope are 100cm and 20 cm respectively. Its magnifying power in normal adjustment is:
- 5
 - 2
 - 25
 - 4
75. An astronomical telescope has an eyepiece of focal length 5 cm. Its angular magnification in normal adjustment is 10, the distance between objective and the eyepiece is:
- 45 cm
 - 50 cm
 - 55 cm
 - 110 cm
76. The least distance of distinct vision is 25 cm. The focal length of a convex lens is 5 cm. It can act as a simple microscope of magnifying power:
- 4
 - 5
 - 6
 - None of these
77. The length of a telescope is 100 cm and magnification is 9. The focal length of the objective and the eyepiece are respectively nearly:
- 90 cm and 10 cm
 - 85 cm and 15 cm
 - 80 cm and 20 cm
 - 95 cm and 5 cm

78. If the focal length of one lens is - 40 cm and that of the other is + 20 cm the focal length of the combination will be:
 a. 80 cm b. 40 cm c. 30 cm d. 35 cm
79. The magnifying power of a simple microscope is 6. The focal length of its length in meters will be: (least distance of distinct vision is 25 cm).
 a. 0.05 b. 0.06 c. 0.25 d. 0.12



- A metal ball-bearing of specific heat c , moving with speed v , is brought to rest. All its kinetic energy is converted into thermal energy which it absorbs, causing a temperature rise ΔT . Where was the value of v ?
 A $\frac{1}{2} c \Delta T$ B $2 c \Delta T$ C $\sqrt{c \Delta T}$ D $\sqrt{2 c \Delta T}$
- The molar heat capacity of chromium at room temperature is $24 \text{ J K}^{-1} \text{ mol}^{-1}$ and its relative atomic mass is 52. What is the specific heat capacity of chromium?
 A $0.46 \text{ J K}^{-1} \text{ Kg}^{-1}$ B $1.2 \text{ J K}^{-1} \text{ Kg}^{-1}$ C $2.2 \text{ J K}^{-1} \text{ Kg}^{-1}$ D $460 \text{ J K}^{-1} \text{ Kg}^{-1}$
- 20 gram of water at 100°C is mixed with 40 gram of water at 00°C , the final temperature of mixture is:
 A 50°C B 6.66°C C 3.33°C D 10°C
- A heat engine has efficiency 25%, the difference between source and sink is 75°C , the temperature of source is:
 A 375 K B 275 K C 300 K D 273 K
- A temperature difference of 25°C is equivalent to a temperature difference of:
 A 25°F B 45°F C 67°F D 77°F
- If the temperature of a patient is 40°C , his temperature on the Fahrenheit scale will be:
 A 72°C B 96°F C 100°F D 104°F
- The correct value of 0°C on the Kelvin Scale is:
 A 273.15 K B 272.85 K C 273 K D 273.2 K
- The gas thermometers are more sensitive than the liquid thermometer because gases:
 A Expand more than liquid C Are much lighter
 B Do not change their state easily D Are easy to obtain
- Device used to measure very high temperature is:
 A Pyrometer B Thermometer C Barometer D Calorimeter
- When water is heated from 0°C to 10°C its volume.
 A First decreases and then increases C Increases
 B Decreases D Remains unchanged
- The coefficient of volume expansion of a solid is x times the coefficient of superficial expansion. Then x is:
 A 3 B 2.5 C 2 D 1.5
- The coefficient of linear expansion of iron is 0.000011 per K. An iron rod is 10 m long at 27°C . The length of the rod will be decreased by 1.1 mm when the temperature of the rod changes to:
 A 0°C B 10°C C 17°C D 20°C
- A metal sheet with a circular hole is heated. The hole will:
 A Contract
 B Expand
 C Remains unaffected
 D Contract or expand depending on the value of the linear expansion coefficient
- Sun's heat reaches us by:
 A Conduction B Radiation C Convection D Scattering

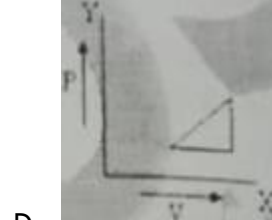
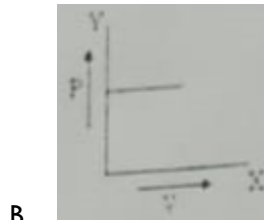
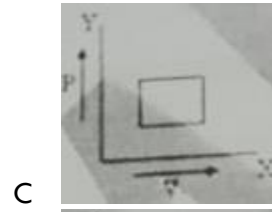
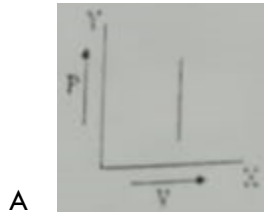
15. The speediest mode of heat transfer is:
 A Conduction B Convection C Radiation D Combustion
16. The high thermal conductivity of metal is due to free electrons. The relevant electron property is:
 A Its being charged C Its high average thermal energy
 B Its high average energy D Its low volume
17. By which of the following methods could a cup of hot tea lose heat, if situated on metal table in a class room.
 A Conduction C Radiation and evaporation
 B Convection D All of these
18. It is hotter at the same distance over the top of a fire that it is on the side of it mainly because:
 A Heat is radiated upwards
 B Air conducts heat upwards
 C Convection takes more heat upward
 D Conduction, convection and radiation all contribute significantly in transferring heat upward
19. According to Wein's displacement Law:
 A $\lambda_m = \text{constant}$ B $\lambda_m T = \text{constant}$ C $\lambda_m T^2 = \text{constant}$ D $\lambda_m^2 T = \text{constant}$
20. The total radiation emitted by a perfectly black body is proportional to:
 A Temperature on ideal gas scale
 B Fourth root of temperature on ideal gas scale
 C Fourth power of temperature on ideal gas scale
 D Square of temperature on ideal gas scale
21. A black body does not:
 A Absorb radiation B Reflect radiation C Refract radiation D B & C Only
22. The SI unit of Stefan's constant is:
 A $\text{Wm}^{-2}\text{K}^{-4}$ B $\text{Nm}^{-2}\text{K}^{-4}$ C $\text{Jm}^{-2}\text{s}^{-1}\text{K}^{-4}$ D $\text{Jm}^{-2}\text{K}^{-4}$
23. If the absolute temperature of a perfectly black body is doubled, then its rate of radiation increases by a factor of:
 A 2 B 4 C 8 D 16
24. A black body at high temperature TK radiates energy at the rate $E = W/\text{m}^2$. When the temperature falls to $\frac{T}{2} K$, the radiated energy in W/m^2 will be:
 A $E/4$ B $E/2$ C $2E$ D $E/16$
25. A hot body will be radiated maximum energy if its surface is:
 A White and polished C Black and rough
 B Black and polished D White and rough
26. A black body at 227°C radiates heat at the rate of $5 \text{ cal}/\text{cm}^2/\text{s}$. The rate of heat radiated in $\text{cal}/\text{cm}^2/\text{s}$ at 727°C is:
 A 40 B 80 C 160 D 240
27. A black body at a temperature of 127°C radiates heat at the rate of $5 \text{ cal}/\text{cm}^2/\text{s}$. At a temperature of 927°C its rate of emission in units of $\text{cal}/\text{cm}^2/\text{s}$ will be:
 A 405 B 35 C 45 D 350

28. If the temperature of a black body increases from 7°C to 287°C , then the rate of energy radiation increases by a factor:
- A $\left(\frac{287}{7}\right)^4$ B 16 C 4 D 2
29. The equation of state corresponding to 8 gm of O_2 is:
- A $PV = 8RT$ B $PV = RT$ C $PV = \frac{RT}{2}$ D $PV = \frac{RT}{4}$
30. A liter of an ideal gas at 27°C is heated at constant pressure to 297°C , what is the approximate final volume?
- A 1.9 liters B 1.2 liters C 19 liters D 0.19 liters
31. Let C_v and C_p denote the molar heat capacities of an ideal gas at constant volume and constant pressure respectively. Which of the following is a universal gas constant?
- A $\frac{C_p}{C_v}$ B $C_p C_v$ C $C_p - C_v$ D $C_p + C_v$
32. The quantity $\frac{PV}{KT}$ represents:
- A Mass of the gas B KE of the gas C Number of molecules in the gas D Number of moles of the gas
33. The ratio of specific heat capacity to molar heat capacity of a body.
- A Is dimensionless B Depends on the molecular weight of the body C Depends on the mass of the body D Is a universal constant
34. If K is the Boltzmann constant, the average KE of a gas molecule at absolute temperature is:
- A $\frac{KT}{2}$ B $\frac{3KT}{4}$ C KT D $\frac{3KT}{2}$
35. The root mean square speed of molecules of a gas at absolute temperature T is proportional to:
- A $1/T$ B T C T^2 D $\sqrt{T}$
36. E_0 and E_h respectively represent the average kinetic energy of a molecule of oxygen and hydrogen. If the two gases are at same temperature, which of the following statements is true?
- A $E_0 = E_h$ B $E_0 > E_h$ C $E_0 < E_h$ D Nothing can be said about the magnitude of E_0 and E_h as the information given is sufficient
37. The average kinetic energy of hydrogen molecules at 300 K is E . At the same temperature, the average kinetic energy of oxygen molecules will be:
- A $E/16$ B E C $E/4$ D $4E$
38. The root mean square speed of gas molecules:
- A Is the same for all gases at the same temperature B Depends on the mass of the gas molecules and its temperature C Is independent of the density and pressure of the gas

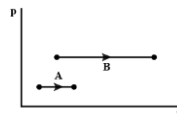
- D Depends only on the temperature and volume of the gas
39. The following four gases are at the same temperature. In which gas do the molecules have maximum root mean square speed?
- A Nitrogen B Oxygen C Hydrogen D Carbon dioxide
40. The temperature of gas is raised from 27°C to 927°C . The root mean square speed of its molecules.
- A Becomes $\sqrt{\frac{927}{27}}$ times the earlier value C Get halved
D Gets doubled
- B Remains the same
41. The temperature of an ideal gas is decreased from 120 K to 30 K. If at 120 K the root mean square velocity of the gas molecules is V , at 30 K it becomes:
- A $4V$ B $2V$ C $V/4$ D $V/2$
42. The pressure and temperature of a given mass of a gas is increased to 3 times each. In this process the rms velocity:
- A Increases to 3 times C Increases to $3\sqrt{3}$ times
D Increases to $\sqrt{3}$ times
- B Increases to 9 times
43. The translational kinetic energy of the molecules for one mole of an ideal gas is equal to:
- A $\frac{3}{2}RT$ B $\frac{2}{3}KT$ C $\frac{1}{2}RT$ D $\frac{2}{3}RT$
44. The heat capacity of a body depends on:
- A The heat given C The material of the body
D Both B & C
- B The mass of the body
45. The relation between pressure and density of an ideal gas is written as:
- A $P = \frac{1}{3}\rho V^2$ B $P = \rho V^2$ C $\rho = \frac{1}{3}PV^2$ D $\rho = PV^2$
46. Metals are good conductors because:
- A Copper B Brass C Aluminum D Silver
47. What does thermal motion mean?
- A Motion due to heat engine C Motion of the body that generates heat
D Random motion of molecules
- B Disorder motion of the body as a whole
48. Thermodynamics mostly deals with:
- A Change of state
B Transfer of heat
C Conversion of heat to other forms of energy
D Measurement of quantity of heat
49. If P is the Pressure, U is the internal energy and dV the volume increase of system, then by definition.
- A $dU = dQ + PdV$ C $dU = dQ - PdV$
D $dU = dQ - VdP$
- B $dU = dQ + VdP$
50. First Law of Thermodynamics represents the conversion of:
- A Heat B Work C Momentum D Energy
51. In a thermodynamic system, if ΔU represents the increase in internal energy and ΔW the work done by the system. Which of the following statements is true?

- A $\Delta U = \Delta W$ in an adiabatic process
 B $\Delta U = -\Delta W$ in an isothermal process
 C $\Delta U = \Delta W$ in an isothermal process
 D $\Delta U = -\Delta W$ in an adiabatic process

52. Which of the following cases, the work done by a gas is minimum?



53. The property of the system that does not change during an adiabatic change in:
 A Temperature B Volume C Pressure D Quantity of heat
54. A curve drawn between two points on P-V diagram represents:
 A Work done in a cyclic process C Work done on or by the system
 B A thermodynamic process D The state of a system
55. In an isothermal expansion of a gas:
 A Temperature remains constant C Volume remains constant
 B Pressure remains constant D Density remains constant
56. If an ideal gas is isothermally expanded, its internal energy will:
 A Remain the same
 B Increase
 C Decrease or increase depending on the nature of the gas
 D Decrease
57. The process on an ideal gas is isothermally expanded, its internal energy will:
 A Isothermal B Isobaric C Isochoric D None of these
58. Consider two processes on a system as shown in figure. The volumes in the initial states are the same in the two processes and the volumes in the final states are also the same. Let ΔW_1 and ΔW_2 be the work done by the system in the process A and B respectively.



- A $\Delta W_1 > \Delta W_2$
 B $\Delta W_1 = \Delta W_2$
 C $\Delta W_1 < \Delta W_2$
 D Nothing can be said about the relation between ΔW_1 and ΔW_2
59. During an adiabatic expansion the increase in volume is associated with:
 A Decrease in pressure and decrease in temperature
 B Increase in pressure and decrease in temperature

- C Increase in pressure and increase in temperature
D Decrease in pressure and increase in temperature
60. If heat is supplied to an ideal gas in an isothermal process:
A The gas will do negative work
B The internal energy of the gas will remain same
C The said process is not possible
D None of these
61. If a system undergoes contraction of volume, then the work done by the system is:
A Zero B Negligible C Positive D Negative
62. If the internal energy of a gas decreases by an amount equal to the external work, the gas is undergoing:
A Adiabatic expansion C Isothermal expansion
B Adiabatic compression D Isochoric expansion
63. Which of the following is the best container of the gas during adiabatic process?
A Glass vessel B Copper vessel C Wood vessel D Thermos flask
64. Thermos flask prevents heat loss by:
A Conduction B Convection C Radiation D All of the above
65. Thermos flask consist of a double walled:
A Glass vessel B Copper vessel C Wood vessel D Iron vessel
66. If a bimetallic strip is heated it will:
A Filled with water C Evacuated
B Filled with H_2SO_4 D None of these
67. A bimetal made of copper and iron strips welded together is straight at room temperature. It is held vertically in the hand so that the iron strip is towards the left hand and copper strip is towards the right hand side. This bimetal is then heated by flame. The bimetal strip will:
A Remain straight C Bend towards left
B Bend towards right D No change
68. When a strip made of iron (having coefficient of linear expansion α_1) and copper (having coefficient of linear expansion α_2 with $\alpha_1 > \alpha_2$) is heated:
A Its length does not change C It bends with iron on concave side
B It gets twisted D It bends with iron on convex side
69. A bimetal made of brass and steel strips welded together is straight at room temperature. It is held vertically in the hand so that the steel strip is towards the left hand and Brass strip is towards the right hand side. The bimetal strip will:
A Remains straight C Bend towards left
B Bend towards right D No change
70. Bimetallic strip use in:
A Breaks in car B Fan C Galvanometer D Thermostat
71. The second law of thermodynamics implies:
A Whole of the heat can be converted into mechanical energy
B No heat engine can be 100% efficient
C Every heat engine has an efficiency of 100%

- D A refrigerator can reduce the temperature to absolute zero
72. The complete conversion of a given quantity of heat into useful mechanical work occurs:
- A Often
 - B Seldom
 - C Never
 - D Depends on the device
73. A refrigerator is:
- A Heat engine
 - B An electric motor
 - C Air cooler
 - D Heat engine working in backward direction
74. The physics underlying the operation of a refrigerator most closely resembles the physics underlying:
- A Heat engine
 - B Melting of ice
 - C The freezing of water
 - D The vaporizing of water
75. A system X is neither in thermal equilibrium with Y nor with Z. The system Y and Z:
- A Must be in thermal equilibrium
 - B Cannot be in thermal equilibrium
 - C May be in thermal equilibrium
 - D None of these
76. A thermodynamic system returns to its original state. Which of the following is/are possible?
- A The work done is zero
 - B The work done is positive
 - C The work done is negative
 - D All of the above
77. What is the name of the device which is used to transfer heat from a sub system at lower temperature to the surroundings at higher temperature?
- A Heat engine
 - B Refrigerator
 - C Carnot's engine
 - D No such device exists
78. Theoretically the efficiency of a Carnot engine is 100% when the temperature of the sink is:
- A 0°F
 - B 0°C
 - C 0°K
 - D 0°R
79. When the temperature difference between the source and the sink increases the efficiency of the heat engine:
- A Increases
 - B Decreases
 - C Is not affected
 - D May increase or decrease depending upon the nature of the working substance
80. If the temperature of a source and sink of a Carnot engine is increased by 10% its efficiency:
- A Increases by 10%
 - B Decreases by 10%
 - C Increases by 1%
 - D Remains the same
81. An ideal heat engine exhausting heat at 27°C is to have 25% efficiency. It must take heat at:
- A 127°C
 - B 227°C
 - C 327°C
 - D 673°C
82. The efficiency of a Carnot engine can be increased by:
- A Increasing both the temperature of source and sink
 - B Decreasing the temperature of source and sink
 - C Decreasing the temperature of source and increasing the temperature of sink
 - D Increasing the temperature of source and decreasing the temperature of sink
83. If the opening door of a refrigerator, which is inside a room:

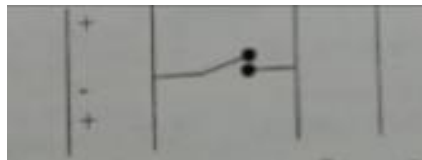
- A You can cool the room the certain degree
 B You can cool it to the temperature inside the refrigerator
 C You ultimately warm the room
 D None of the above
84. If the heat is removed from a body, the change in entropy is:
 A Positive B Negative C Remains constant D None of these
85. If heat is entered in the system, the change in entropy is:
 A Positive B Negative C Remains constant D None of theses



CHAPTER 12: ELECTROSTATISTICS

ELECTROSTATISTICS

- Two parallel metallic plates connected with a 12-volt battery are separated by 10 cm, intensity between plate is:
 - 1.2 J/C
 - 120 J/C
 - 12 N-m
 - 10 N-m
- An electron is accelerated by volts, its kinetic energy in electron volt is:
 - 1.6×10^{-19} eV
 - 1.6×10^{-19} eV
 - 10 eV
 - None of these
- K.E. of an electron accelerated through a potential difference of 1000 volts will be:
 - 16×10^{16} J
 - 1.6×10^{16} J
 - 16×10^{-16} J
 - 1.6×10^{-13} J
- If $4 \mu\text{F}$ and $2 \mu\text{F}$ capacitors are connected in series, the equivalent capacitor is:
 - $0.7 \mu\text{F}$
 - $6 \mu\text{F}$
 - $2 \mu\text{F}$
 - $1.33 \mu\text{F}$
- A dielectric = 2 is inserted between the plates of a $20 \mu\text{F}$ capacitor. Its capacitance will become:
 - $10 \mu\text{F}$
 - $18 \mu\text{F}$
 - $22 \mu\text{F}$
 - $40 \mu\text{F}$
- Capacitor A has charge q on it whereas B is uncharged. When the switch is ON, the charge appearing on B is:



- Zero
 - $-q/2$
 - q
 - $2q$
- A gold nucleus (radius r) is represented by the symbol ${}_{79}^{197}\text{Au}$. Taking as the elementary charge and ϵ_0 as the permittivity of free space, what is the electric field strength at the surface of an isolated gold nucleus?
 - Zero
 - $\frac{79e}{4\pi\epsilon_0 r^2}$
 - $\frac{197e}{4\pi\epsilon_0 r^2}$
 - $\frac{197e^2}{4\pi\epsilon_0 r^2}$
 - At what distance from an electric charge of 10 coulombs, the electric field intensity be zero?
 - 10 m
 - 0 m
 - Infinity
 - None of them
 - The electric flux due to an electric dipole is
 - Gauss's law holds only for closed surfaces.
 - Electric field intensity due to an infinite sheet of charge is $E = \frac{\sigma}{2\epsilon_0}$.
 - Electric field intensity between two oppositely charged plates is $E = \sigma/\epsilon_0$.
 - Where σ = charge density (charge per unit area)
 - The potential difference in moving a test charge along an Equipotential surface is:
 - Maximum
 - Zero
 - $\frac{kq}{r}$
 - None of these
 - The charge on the uranium nucleus is 1.5×10^{-17} C and the charge on the α - particle is 3.2×10^{-13} C. What is the electrostatic force between a uranium nucleus and α - partial separated by 1.0×10^{-13} m?
 - 4.3×10^{-33} N
 - 4.3×10^{-20} N
 - 4.3×10^{-43} N
 - 4.3 N
 - 4.3×10^{30}

12. Decreasing the separation of two positive charges by one half will cause the force of repulsion to be changed by:
- $\frac{1}{4}$ times
 - 2 times
 - $\frac{1}{2}$ times
 - 4 times
13. The electrostatic force between two charges is F, how much distance between two charges be changed to be force 2F.
- $\frac{1}{4}$ times
 - 2 times
 - $\frac{1}{2}$ times
 - $\frac{1}{\sqrt{2}}$ times
14. The force between two charges is 28N. if the paraffin wax of relative permittivity 2.8 is introduced between the charges as a medium, then the force reduces to:
- 25 N
 - 20 N
 - 15 N
 - 10 N
15. Branch of physics that deals with the study of charges at rest is called:
- Electrostatics
 - Electrodynamics
 - Electromagnetism
 - Modern physics
16. The laws of forces that govern the force between two electric charges were discovered by:
- Faraday
 - Ampere
 - Coulomb
 - Ohm
17. Coulomb's law is given by $F = kq_1q_2r^n$ where, n is:
- 1/2
 - 2
 - 2
 - 1/2
18. Coulomb's law the force:
- Between two hard metallic spheres
 - Between two billiard balls
 - Between two-point charges
 - None of these
19. Coulomb's force between two-point charges depends upon:
- The amount of charges on them
 - The distance between them
 - The medium in which they are placed
 - All of the above
20. The S.I. unit of electric permittivity is:
- Nm^2C^{-2}
 - Am^{-1}
 - Nc^{-1}
 - $\text{C}^2\text{N}^{-1}\text{m}^{-2}$
21. The force between two charges each of 1 μC placed at a distance of 1 meter in air is
- $9 \times 10^{-12} \text{ N}$
 - $9 \times 10^{-9} \text{ N}$
 - $9 \times 10^9 \text{ N}$
 - None of these
22. The Coulomb's force between two charges in a medium of relative permittivity ϵ_r is:
- $F_1 = \epsilon_r F$
 - $F_1 = \epsilon_r + F$
 - $F_1 = \frac{F}{\epsilon_r}$
 - None of these
23. In the distance between the two charge bodies is doubled, the Coulomb's force between them becomes:
- Four times
 - One fourth
 - Half
 - Double
24. S.I. unit of charge is:
- Joule
 - Volt
 - Ampere
 - Coulomb
25. Force experienced by a unit positive charge at a point in the electric field is called:
- Electric current
 - Electric field intensity
 - Electric flux
 - Electric potential
26. S.I. unit of electric (field) intensity is:

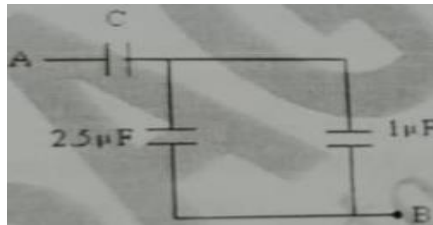
- a. Newton-coulomb
b. Newton/coulomb
27. An electric charge at rest produces:
a. Only an electric field
b. Only a magnetic field
c. Neither electric nor magnetic field
d. Both electric and magnetic field
28. Electric lines of forces about a negative charge are:
a. Circular, anticlockwise
b. Circular clockwise
c. Radial, inward
d. Radial, outward
29. The dot product of electric intensity and vector area is called:
a. Electric potential
b. Electric intensity
c. Electric forces
d. Electric flux
30. Electric flux is:
a. A vector quantity
b. A scalar quantity
c. Vector as well as scalar quantity
d. None of these
31. Electric flux through any surface depends upon:
a. The electric intensity
b. The area of the surface
c. Direction of surface relative to the field
d. All of these above
32. Electric flux per unit area is a measure of:
a. Electric force
b. Electric intensity
c. Electric potential
d. None of these
33. A suitable unit for expressing electric field strength is:
a. N/C
b. Am
c. C/m²
d. V/C
34. A force of 0.01 N is exerted on a charge of 1.2×10^{-2} C at a certain point. The electric field at that point is:
a. 5×10^4 N/C
b. 5.3×10^2 N/C
c. 8.3×10^2 N/C
d. 8.3×10^4 N/C
35. Two thin infinite parallel plates have uniform charge densities $+\sigma$ and $-\sigma$. The electric field in the space between them is:
a. $\sigma/2E_0$
b. σ/E_0
c. $2\sigma/E_0$
d. Zero
36. Gauss's law can only be applied to:
a. A flat surface
b. A curved surface
c. A closed surface
d. A surface of any shape
37. "The flux through any closed surface is $1/E_0$ times the total charge enclosed in it" represents the statement of:
a) Gauss's law
b) Coulomb's law
c) Newton's law of gravity
d) Stoke's law
38. The work done in moving a unit positive charge against the electric field is called:
a) electric force
b) electrical potential
c) electric flux
d) none of these
39. Which of the following is a volt?
a. Erg per cm
b. Joule coulomb
c. Erg per ampere
d. Joule per coulomb
40. An electron initially at rest is accelerated through a P D of one volt. The energy acquired by electron is:

- a. 1J
b. 1.6×10^{-10} erg
c. 10^{-19} J
d. 1.6×10^{-19} J
41. How many electrons will have a charge of one Coulomb?
a. 6.25×10^{18} e b. 6.25×10^{19} e c. 5.25×10^{18} e d. 5.25×10^{19} e
42. An electron of mass 'm' and charge 'e' travels from rest through a potential difference V. its final energy is:
a. mev b. ev c. e/v d. ev/m
43. The relation between electric V and electric intensity E is:
a. $V = E/d$ b. $V = Ed$ c. $Vd = E$ d. $V = Ed^2$
44. An electron volt is equal to:
a. 0.62×10^{13} J b. 16×10^{-13} J c. 0.62×10^{19} J d. 1.6×10^{19} J
45. Work done in moving a unit positive charge through a distance of 'x' meter on an equipotential surface is:
a. X Joule b. 1/X Joule c. Zero d. X^2 Joule
46. If a unit charge, I taken from one point to another over an equipotential surface, then
a. work is done on the charge c. work done on the charge is constant
b. work is done by the charge d. no work is done
47. The work done in moving a positive charge on an equipotential surface is:
a. Finite and positive c. finite and negative
b. Infinite d. Zero
48. No current flows between two charged bodies when connected:
a. if they have some capacity
b. if they have the same quantity of charge
c. if they have same potential
d. if they have the same charge-to-potential ratio
49. An electron of mass 'm' and charge 'e' is accelerated from rest through a potential difference V in vacuum. Its final speed will be:
a. $\sqrt{\frac{2eV}{m}}$ b. $\sqrt{\frac{eV}{m}}$
c. $eV/2m$ d. eV/m
50. Electron volt (ev) is a unit of:
a. Energy b. Potential c. Current d. Charge
51. The electric potential near an isolated point charge is:
a. $K \frac{q}{r^2}$ c. $K \frac{q^2}{r}$
b. $K \frac{q}{r}$ d. $K \frac{q^2}{r^2}$
52. Equal charges are given to two spheres of different radii. The potential will be:
a. More on smaller sphere
b. More on bigger sphere
c. Equal on both the spheres
d. Dependent on nature of material of the spheres
53. Value of potential at a point due to a point charge is:

- a. inversely proportional to square of the distance
 - b. directly proportional to square of the distance
 - c. inversely proportional to the distance
 - d. directly proportional to the distance
54. The electric flux through the surface of a sphere having a point charge at its center depends upon:
- a. Surface area of sphere
 - b. Radius of sphere
 - c. Quantity of charge inside the sphere
 - d. None of these
55. A device used to store charge is called:
- a. Capacitor
 - b. Photocell
 - c. Ammeter
 - d. Galvanometer
56. The unit of capacitance is:
- a. Coulomb
 - b. Ampere
 - c. Volt
 - d. Farad
57. The capacitance unit of convenient size is:
- a. Farad
 - b. Micro Farad
 - c. Kilo Farad
 - d. Mega Farad
58. Two capacitors of capacities C_1 and C_2 are connected in parallel. If a charge Q is given to the assembly, it gets shared. Ratio of the charge on capacitor C_1 to the charge on capacitor C_2 is given by:
- a. $\frac{C_1}{C_2}$
 - b. $\frac{C_2}{C_1}$
 - c. $\frac{C_1^2}{C_2^2}$
 - d. $\frac{C_2^2}{C_1^2}$
59. Capacitor connected in series have:
- a. the same difference of potential across each capacitor
 - b. the effective capacitance equal to the sum of the individual capacitance
 - c. numerically the same charge on each plate of all the capacitor
 - d. none of the above
60. Three capacitor of capacitance $2\mu\text{F}$ each are connected in parallel. Their equivalent capacitance is:
- a. $1\mu\text{F}$
 - b. $2.5\mu\text{F}$
 - c. $6\mu\text{F}$
 - d. $8\mu\text{F}$
61. Three capacitor of capacitance $3\mu\text{F}$ each are connected in series. Their equivalent capacitance is:
- a. $1\mu\text{F}$
 - b. $6\mu\text{F}$
 - c. $3\mu\text{F}$
 - d. $1\mu\text{F}$
62. Three capacitors or capacitance $4\mu\text{F}$, $6\mu\text{F}$ and $12\mu\text{F}$ are connected first on series and then in parallel. What the ratio of equivalent capacitance in the two cases?
- a. 2:3
 - b. 1:11
 - c. 11:1
 - d. 1:3
63. The equivalent capacitance of three capacitors of capacitance C_1 , C_2 and C_3 are connected in parallel is 12 units and the product $C_1 C_2 C_3 = 48$. When the capacitors C_1 and C_2 are connected in parallel the equivalent capacitance is 6 units. The capacitance are:
- a. 2, 4, 6
 - b. 1, 5, 6
 - c. 1.5, 2.5, 8
 - d. 2, 3, 7
64. Three different capacitors are connected in series then:
- a. they will have equal charges
 - b. they will have same potential
 - c. Both a and b
 - d. None of these
65. Two capacitors of $3\mu\text{F}$ and $6\mu\text{F}$ are connected in series across a potential difference of 120 volt. Then the potential difference across $3\mu\text{F}$ capacitor is:

- a. 40 V b. 60 V c. 80 V d. 100 V
66. When the capacitance are combined in series, the quality which remains constant is:
a) intensity of electric field b) charge
c) potential difference d) energy
67. The electric field between the plates of a parallel plate capacitor is E. What is the field between the plates after immersing the capacitor in a liquid of relative permittivity 2?
a. 2E b. $\frac{E}{\sqrt{2}}$ c. $\sqrt{2}E$ d. $\frac{E}{2}$
68. The capacitance of parallel plates capacitor is:
a. $\frac{d \epsilon_0}{A}$ c. $\frac{Ad}{\epsilon_0}$
b. $\frac{\epsilon_0}{Ad}$ d. $\frac{A\epsilon_0}{d}$
69. Two capacitance $4\mu F$ and $6\mu F$ are connected in series. A potential difference of 500 V is applied between the outer plates of the system. The charge on each capacitor will be:
a. 1200 C b. 6000 C c. $1200\mu C$ d. $6000\mu C$
70. When any medium between the plates is used as a dielectric, the capacitance of capacitor is:
a. $\frac{A \epsilon_0 \epsilon_r}{d}$ c. $\frac{A \epsilon_r}{d \epsilon_0}$
b. $\frac{A \epsilon_0}{d \epsilon_r}$ d. $\frac{\epsilon_0 \epsilon_r}{Ad}$
71. A parallel plate capacitor has capacitance of 50PF in air and 105 PF when immersed in oil. The dielectric constant of the oil is:
a. 50/105 b. 1 c. 21 d. ∞
72. energy of an electrical condenser of capacity 'c', when subjected to a potential 'v' is given by
a. $1/2 cv^2$ c. cv
b. $1/2 c^2 v$ d. $\frac{1}{2} \frac{c}{v}$
73. if a $2\mu F$ condenser is charged to 200 volts and its plates are connected by a wire, then the heat produced in the wire is:
a. $4 \times 10^{-4} J$ b. $4 \times 10^{-2} J$ c. $4 \times 10^{-5} J$ d. $2.5 \times 10^{-2} J$
74. A microfarad capacitor is to have an energy content of 50J in order to operate a flesh lamp. The voltage required to charge the capacitor is:
a. 500 V b. 1000 V c. 1500 V d. 2000 V
75. A parallel plate capacitor with air between the plates has a capacitance of 4PF. What will be the capacitance if the distance between the plates is reduced by half and the space between them is filled with a substance of dielectric constant 6?
a. 6 PF b. 12 PF c. 24 PF d. 48 PF
76. A condenser of $50\mu F$ is charged to 10 volts. Its energy in joules is:
a. 2.5×10^{-3} b. 5×10^{-3} c. 10×10^{-4} d. 2.5×10^{-4}
77. Three capacitors each of $1\mu F$, are connected in parallel. To this combination, a fourth capacitor of $1\mu F$ is connected in series. The equivalent capacitance of the system is:
a. $3/4\mu F$ b. $4/3\mu F$ c. $2\mu F$ d. $4\mu F$

78. The equivalent capacitance between A and B in the figure is $1 \mu\text{F}$. Then the value of the capacitance C is:



- a. $1.4 \mu\text{F}$ b. $2.5 \mu\text{F}$ c. $3.5 \mu\text{F}$ d. $1.2 \mu\text{F}$
79. A capacitor remains connected to a battery, a dielectric slab is applied between the plates. The energy will increase due to:
- a. Increase of potential difference c. Increase of capacitance
b. Increase of electric field strength d. None of these
80. The capacitance of a capacitors does not depend on:
- a. The shape of the plates c. The charges on the plates
b. The size of the plates d. The separation between the plates
81. The space between the plates of a capacitor is filled by a liquid of dielectric constant. The capacitance of the capacitor is:
- a. Increase by a factor K c. Decrease by a factor K
b. Increase by a factor K^2 d. Decrease by a factor K^2
82. Increase the charge on the plates of a capacitor means:
- a. Increasing the capacitance
b. Increasing the potential difference between the plates
c. Both a and b
d. None of these
83. The capacitance of a capacitor depends on:
- a. The shape of the plates c. The separation between the plates
b. The size of the plates d. All of the above

CHAPTER 13: CURRENT ELECTRICITY

MULTIPLE CHOICE QUESTIONS

1. The filament of a 240 V, 100W electric lamp heats up from room temperature to its operating temperature. As it heats up, its resistance increases by a factor of 16. What is the resistance of this lamp at room temperature?
 A. 36Ω B. 580Ω C. 1.5Ω D. $9.2K \Omega$
2. A generator produces 100kw of power at a potential difference of 10kv. The power is transmitted through cables of total resistance 5Ω . How much power is dissipated in cables?
 A. 50W B. 250W C. 500W D. 1000W
3. An electric kettle of 1000 watts boils certain quantity of water in 480 seconds. The heat has been generated to boil this water will be:
 A. 42×10^4 Joules C. 48×10^6 Joules
 B. 48×10^4 Joules D. 48×10^6 Joules
 E. None of the above
4. A storage whose emf is 12v and whose internal resistance is 0.2Ω is to be charge at rate of 20A. So the applied voltage should be:
 A. 8V C. 16V
 B. 24V D. None of the above
5. A battery whose e. m. f. is 40V has resistance of 5Ω . If this battery is connected to a 5Ω resistor 'R', what will be the voltage drop across 'R'?
 A. 10 V C. 40 V
 B. 30 V D. 50 V
 E. 70 V
6. A current of 4.8 A is flowing in a conductor. The number of electrons passing through any cross section per second is:
 A. 3×10^{19} B. 3×10^{20} C. 7.68×10^{19} D. 7.68×10^{20}
7. A current of 16m A flows through a conductor. The number of electrons flowing per second is
 A. 10^{14} B. 10^{15} C. 10^{-17} D. 10^{17}
8. The current carried through copper wire is due to the motion of:
 A. Neutron C. Protons
 B. Free electrons D. Protons and Neutrons
9. In an external circuit the direction of flow of conventional current is
 A. In the direction of flow of electrons C. From positive to negative
 B. From negative to positive D. None of the these
10. Which of the following is correct?
 A. 1 ampere = 1 coulomb $\times$ 1 second C. 1 ampere = 1 coulomb/1 second
 B. 1 ampere = 1 second/1coulomb D. None of these
11. The S.I unit of current is:
 A. Coulomb B. Coulomb-second C. Volt D. Ampere
12. If a net charge Q flows across any cross section of the conductor in time t, then current I through it is given by:
 A. $I = Qt$ B. $I = t/Q$ C. $I = Q/t$ D. $I = 2t/Q$

13. If a charge of 60 coulomb flows through the cross section of a conductor in one minute, the current flowing through it is
 A 60 A B 3600 A C 30 A D 1 A
14. Which of the following represents an electric current?
 A J/sec B Erg/sec C N/sec D C/sec
15. If a charge of 180 coulomb flows through the cross section of a conductor in one minute, the current flowing through it is:
 A 1 A B 3 A C 5 A D 7 A
16. If the temperature of the conductor (metal) is kept constant, then the current flowing through it is:
 A Inversely proportional to the applied voltage
 B Directly proportional to the applied voltage
 C Independent of the applied voltage
 D All of the above
17. Which of the following represents Ohm's law?
 A $I = VR$ B $V = IR$ C $R = IV$ D $V = I/R$
18. For a fixed resistance, if the voltage applied is doubled the current will
 A Remain the same C Be doubled
 B Become half D Become one fourth
19. If the current through a fixed resistance is halved then:
 A The potential difference is halved C The resistance is halved
 B The power is halved D None of these
20. Electrical resistance is the:
 A Opposition to the flow of electrons
 B Support to the flow of electron
 C Neither support nor opposition to the flow of electrons
 D Additional
21. The resistance of a conductor depends upon
 A The magnitude of current passing through it
 B The potential difference between its ends
 C Its physical state dimension and nature of its material
 D All of the above
22. The resistance of a conductor:
 A Increase with increase in its length C Increases with increase in temperature
 B Decrease with increase in area D All of the above
23. The unit resistance "Ohm" is defined as:
 A Ampere/Coulomb C Volt/Ampere
 B Ampere/Volt D Volt/Coulomb
24. On increasing the temperature of a conductor, its resistance:
 A Decreases C Remains unchanged
 B Increases D Volt/Coulomb

25. The resistance R_0 and R_t of a metallic wire at temperature 0°C and $t^\circ\text{C}$ are related as (α is temperature coefficient of resistance)
- A $R_t = R_0 (1 + \alpha t)$ C $R_t = R_0^2 (1 + \alpha t)$
 B $R_t = R_0 (1 - \alpha t)$ D $R_t = R_0 (1 - \alpha t)$
26. The reciprocal of resistance is:
- A Conductance B Resistivity C Conductivity D None of these
27. Which of the following statements is not true?
- A Conductance is the reciprocal of resistance and is measured in Siemens
 B Ohm's Law is not applicable at very low and very high temperature
 C Ohm's Law is applicable to semiconductors
 D Ohm's Law is not applicable to electron tubes, discharge tubes and electrolysis
28. The resistance of a material increase with temperature. It is a:
- A Metal B Insulator C Semiconductor D Semi metal
29. The resistivity of a material having length L , resistance R and area of cross section A .
- A $\rho = RLA$ B $\rho = RL/A$ C $\rho = LA/R$ D $\rho = RL/A$
30. The resistivity of the material of a conductor depends on:
- A Its length C Its area of cross section
 B Its resistance D None
31. The SI unit of resistivity is:
- A Ohm B Ohm/m C Ohm-m D Ohm-m²
32. If the length of a wire is doubled, then its resistance will
- A Becomes four times C Becomes two times
 B Remains unchanged D Becomes one fourth
33. The resistance of a wire of uniform diameter d and length L is R . The resistance of another wire of the same material but diameter $2d$ and length $4L$ would be:
- A $R/4$ B $R/2$ C R D $2R$
34. The conductivity of a super conductor is:
- A Infinite B Very large C Very small D Zero
35. The resistance of a wire is R . It is stretched uniformly so that its length is tripled. The new resistance of the wire:
- A $3R$ B R C $R/3$ D $9R$
36. The equivalent resistance of a series combination of resistance is:
- A Lower than the highest resistance C Lower than the lowest resistance
 B Higher than the highest resistance D Higher than the lowest resistance
37. The equivalent resistance of a parallel combination of resistance is:
- A Lower than the highest resistance C Lower than the lowest resistance
 B Higher than the highest resistance D Higher than the lowest resistance
38. Two unequal resistances are connected in parallel across a battery. Which of the following statement is true?
- A Same current will flow through both resistances
 B Current through smaller resistances is higher
 C Current through larger resistances is higher

- D Current can be higher in any resistance depending upon emf of the cell
39. If three resistances R_1 , R_2 and R_3 are joined in parallel across some potential source, then:
- Equivalent resistance < Lowest individual resistance
 - Applied potential difference = potential difference across each resistance
 - Current supplied by the source is shared among three resistances inversely in their ratio
 - All of the above
40. How many times will the resistance of "n" identical conductors be increased if the parallel resistance be changed to series one?
- $\sqrt{n}$
 - n
 - n^2
 - n^{-2}
41. Two resistances R_1 and R_2 have effective resistance R_s when connected in series and R_p when connected in parallel. If R_s and R_p and $\frac{R_1}{R_2} = 4$. Calculate the value of R_1 and R_2 .
- $R_1 = 1, R_2 = 0.25$
 - $R_1 = 2, R_2 = 0.5$
 - $R_1 = 4, R_2 = 1$
 - $R_1 = 8, R_2 = 2$
42. Three resistors of 2 ohm, 3 ohm and 5 ohm are connected in parallel across a battery of 10 V and of negligible internal resistance. The PD across the 3 ohm resistor is:
- 2V
 - 3V
 - 9V
 - 10V
43. In order to obtain maximum resistance from some given resistors, they should be joined in:
- Parallel
 - Series
 - Mixture of series and parallel combinations
 - None of these
44. As the temperature of a conductor increases, its resistivity and conductivity change. The ratio of resistivity to conductivity.
- Increases
 - Decreases
 - Remain constant
 - May increase or decrease depending on the actual temperature
45. An electric iron is rated at 240 V, 480 W. The current through it will be:
- 240 A
 - 480 A
 - 2 A
 - 0.5 A
46. The power rating of an electric motor that draws a current of 3A when operated at 240 volt is:
- 720 W
 - 240 W
 - 80 W
 - Zero
47. If R_1 and R_2 are the filament resistance 200 Watt bulb respectively, both designed to run at the same voltage, then:
- R_1 is four times R_2
 - R_2 is four times R_1
 - R_1 is two times R_2
 - R_2 is two times R_1
48. Two bulbs in a house, one glows brighter than the other. Which of the two has a large resistance.
- The dim bulb
 - The bright bulb
 - Both have the same resistance
 - The brightness does not depend upon the resistance
49. Domestic appliances (TV, Radio, Bulb, Refrigerator etc.) are connected:
- In parallel to each other
 - In series to each other

- C Low power appliances in parallel, high power appliances in series
D High power appliances in parallel, low power appliances in series
50. Electromotive force and terminal potential difference are both measured in:
A Ampere B Coulomb C Volt D Farad
51. Electromotive force is most closely related to:
A Electric field C Potential difference
B Magnetic field D Mechanical force
52. If n identical cells, each of emf is E , are connected in series then the net emf is:
A E/n B nE C n^2E D None of these
53. If n identical cells, each of emf is E , are joined in parallel, the net emf is:
A E B E/n C E/n^2 D n^2E
54. If the terminal potential difference of a battery of emf " E " is " V ", its internal resistance is given by:
A $\frac{E-V}{I}$ B $\frac{E+V}{I}$ C $\frac{EI}{V}$ D $E-Ir$
55. A car battery has emf 12 volts and internal resistance 5×10^{-2} ohm. If it draws 60 amp current, the terminal voltage of the battery will be:
A 15 volt B 3 volt C 5 volt D 9 volt
56. A current of 2.0 A passes through a cell of emf is 1.5 V, having internal resistance of 0.15 ohm. The potential difference, in volts across the ends of the cell will be:
A 1.35 B 1.50 C 1.00 D 1.20
57. Four cells of equal emf (E) and internal resistance (r) are connected in series. If one cell is connected wrongly, the equivalent emf and internal resistance will be:
A $3E$ and $3r$ B $2E$ and $4r$ C $2E$ and $3r$ D $3E$ and $4r$
58. Dry cell supplies:
A Alternating current only C Both direct as well as alternating current
B Direct current only D None of the above

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DIETS OF DICC

MULTIPLE CHOICE QUESTIONS

1. If a magnetic induction has a value of 10 Webers/m² at a certain point in a magnetic field, the flux density is equal to:

A 10 Webers/m ²	C All of the above
B 1000 Webers/m ²	D None of the above
2. A 5 m wire carrying a current of 2A is at right angles to the uniform magnetic field of 0.5 Webers/m², the force on the wire is:

A 2 N	B 4 N	C 5 N	D 2.5 N
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3. Due to the magnetic force, a positively charged particle executes uniform circular motion within a uniform magnetic field 'B'. If the charge is 'q' and the radius of its path is 'r', which of the following expressions gives the magnitude of the particle's linear momentum?

A qBr	B $\frac{qB}{r}$
C $\frac{q}{(Br)}$	D $\frac{B}{(qr)}$
E $\frac{r}{(qB)}$	
4. A beam of electron is fired through the axis of current carrying solenoid, the speed of electrons will:

A Increase	B Decrease	C Not change	D Zero
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5. A magnetic field of B is produced in an air core toroid, if core is replaced by iron, whose permeability is 300 times the permeability of air, the value of B:

A Decreases by 300 times	C Not change
B Increases by 300 times	D Zero
6. What is the self-inductance of a coil which produces 5V when the current changes from 3 to 2A in 1 ms?

A 5000 Henry	B 5 milli Henry	C 50 Henry	D 5 Henry
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7. The magnetic effect of an electric current was discovered by:

A Flemming	C Ampere	D Oersted
B Faraday		
8. A charged particle is projected into a uniform magnetic field with its velocity perpendicular to the direction of the field. Which of the following quantities of the particle will not change.

A Momentum	C Velocity
B Speed	D None of the above
9. Induced emf produced in a coil rotating in a magnetic field will maximum when the angle between the axis of coil and direction of magnetic field is:

A 0°	B 90°	C 45°	D 180°
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10. An electron enters region where magnetic field B and electric field E are mutually perpendicular to one another.

A It will always move in the direction of E	C It always possesses circular motion
B It will always move in the direction of B	D It can go undeflected also
11. Moving electron charges will interact with:

A Electric field only	C Both of these
B Magnetic field only	D None of these

12. A proton enters in a magnetic field of B Weber/ m^2 making an angle of 30° with the direction of magnetic field with velocity V . Lorentz force acting on the portion is:
 A eVB B Zero C ∞ D $eVB/2$
13. An electron moves with a velocity V in x -direction. If a magnetic field acts on it in y -direction, the force on the electron acts in:
 A x -direction C z -direction
 B y -direction D Arbitrary direction
14. Magnetic field is not associated with:
 A A charge in uniform motion C A decelerated charge
 B An accelerated charge D A stationary charge
15. A current (I) carrying conductor length (l) is placed in a magnetic field (B). The maximum force acting on the conductor is:
 A IBl B IB/l C I/Bl D B/lI
16. The radius of curvature of the path of a charged particle in a uniform magnetic field is directly proportional to:
 A The charge on the particle C The energy of the particle
 B The momentum of the particle D The intensity of the field
17. One Weber/ m^2 is equal to:
 A 10^4 Gauss B 10^2 Gauss C 10^{-2} Gauss D 10^{-4} Gauss
18. A magnetic needle is kept in a non-uniform magnetic field. It experiences:
 A A force and a torque C A force but not a torque
 B A force but not a torque D Neither a force nor a torque
19. Magnetic lines of force:
 A Intersect at infinity C Intersect with the magnet
 B Cannot intersect at all D Intersect at the neutral points
20. A charged particle of charge q moving with velocity V enters along the axis of a current carrying solenoid. The magnetic force on the particle is:
 A 0 C Finite but not qVB
 B qVB D Infinite
21. Two infinitely long parallel wires separated by a distance b are carrying a current I amp each. The magnitude of the force per unit length exerted by one wire on the other is:
 A $\frac{\mu_0 I^2}{b^2}$ B $\frac{\mu_0 I^2}{2\pi b}$ C $\frac{\mu_0 I}{2\pi b}$ D $\frac{\mu_0 I^2}{2\pi b^2}$
22. The permeability of free space, in $\frac{\text{Weber}}{\text{Ampere} \times \text{Meter}}$ is:
 A $\frac{1}{4\pi \times 10^{-7}}$ B $\frac{1}{4\pi \times 10^7}$ C $4\pi \times 10^{-7}$ D $4\pi \times 10^7$
23. Magnetic field inside a solenoid is:
 A Directly proportional to current
 B Inversely proportional to current
 C Directly proportional to its length
 D Inversely proportional to the total number of turns

24. The magnetic field in the middle of a solenoid are:
 A Circles C Parallel to the axis
 B Spirals D Perpendicular to the axis
25. The self-inductance of a straight conductor is:
 A Zero B Infinity C Very large D Very small
26. Lenz's Law is a consequence of the law of conservation of:
 A Charge B Mass C Energy D Momentum
27. The self-inductance of a coil is a measure of:
 A Electrical inertia B Electrical friction C Induced emf D Induced current
28. The phenomenon of electromagnetic induction was discovered by:
 A Flemming B Oersted C Faraday D Henry
29. An electric motor:
 A Creates electrical energy
 B Creates mechanical energy
 C Converts mechanical energy into electrical energy
 D Converts electrical energy into mechanical energy
30. When a number of turns in a coil is double without any change in the length of the coil, its self-inductance becomes:
 A Four times B Double C Halved D Squared
31. Lenz's Law gives:
 A Direction magnetic field
 B Direction of motion of conductor
 C Direction of induced current
 D Direction of current in any electrical circuit
32. Henry is equal to:
 A 1 (Weber/Ampere) C 1 Weber-Ampere
 B 1 (Weber/Volt) D 1 Weber-Volt
33. The mutual inductance of a pair of coils is 2H. If the current in one of the coils changes from 10A to zero in 0.1 s, the emf induced in the other coil is:
 A 20 V B 200 V C 0.2 V D 200 V
34. Dynamo is based on the principle of:
 A Electromagnetic induction C Mutual induction
 B Self-induction D None of the above
35. A dynamo:
 A Generates mechanical energy
 B Generates electrical energy into mechanical energy
 C Converts electrical energy into mechanical energy
 D Converts mechanical energy into electrical energy
36. To convert mechanical energy into electrical energy, one can use:
 A AC Dynamo B Transformer C DC Dynamo D A & C Both

37. A transformer is used to convert:
- Mechanical energy into electrical energy
 - Alternating current into direct current
 - Direct current into alternating current
 - AC of high voltage and low amperage into AC of low voltage and high amperage and vice versa
38. In a step down transformer the input voltage is 22 KV and the output voltage is 550 V. The ratio of the number of turns in the secondary to that in the primary is:
- 1:20
 - 20:1
 - 1:40
 - 40:1
39. In a noiseless transformer and alternating current of 2A is flowing in the primary coil. The number of turns in the primary and secondary coils are 100 and 20 respectively. The value of the current in the secondary coil is:
- 0.08 A
 - 10 A
 - 0.4 A
 - 5 A
40. An ideal transformer is used to step up an alternating of emf of 220 V to 4.4 KV to transmit 6.6 KW of power. The current of the secondary coil is:
- 30 A
 - 3 A
 - 1.5 A
 - 1 A
41. In a transformer, the number of turns in primary and secondary coils are 1000 and 3000 respectively. If the primary is connected across 80 V AC, the potential difference across each turn of the secondary will be:
- 240 V
 - 0.24 V
 - 0.8 V
 - 0.08 V
42. A step up transformer is used on a 120 V line to provide a potential difference of 2400 V. If the primary coil has 75 turns, the number of turns in the secondary coil is:
- 1500
 - 1200
 - 150
 - 1575
43. In an LCR circuit, the energy is dissipated in:
- R only
 - R and L only
 - R and C only
 - R, L and C only
44. The impedance of a circuit containing inductance L and resistance R is given by:
- LR
 - L/R
 - $\sqrt{R^2 + \omega^2 L^2}$
 - $\sqrt{\omega^2 R^2 + L^2}$
45. The dimension of R is the same as that of L/R.
- True
 - False
 - Both false and true
 - Neither true nor false
46. L, C, R represents the physical quantities inductance, capacitance and resistance respectively. The combinations which have the dimensions of frequency are:
- C/L
 - RL/C
 - R/L
 - None of the above
47. At resonance, in a series LCR circuit, which relation does not hold.
- $\Omega = 1/LC$
 - $\omega = 1/\sqrt{LC}$
 - $L\omega = 1/LC$
 - $C\omega = 1/L\omega$
48. A proton moving with a velocity of 10^4 m/s enters a magnetic field first at 45° and then at 90° to the field. The forces acting on the proton in two cases are in the ratio.
- $\sqrt{2}:1$
 - $1:\sqrt{2}$
 - $1:\sqrt{3}$
 - $\sqrt{3}:1$

49. An electron moving with a velocity of 2×10^5 m/s enters a magnetic field at an angle of 30° to it. Another electron moving at 10^5 m/s enters the same magnetic field at 60° to it. The forces acting on electron are in the ratio.

A $2:\sqrt{3}$

B $\sqrt{3}:2$

C $1:\sqrt{3}$

D $\sqrt{3}:1$

